Testing two-component treatment effects in truncated continuous outcomes

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Abstract

Continuous outcomes such as quality-of-life scores are often undefined after death or a related event. A common pragmatic solution is to encode such patients by the worst possible score and analyze the resulting composite outcome with a one-dimensional procedure. That preserves a clinically meaningful ordering, but it conflates treatment effects on the atom with treatment effects on the continuous part among those with observed outcomes. We propose likelihood-ratio tests for the joint null of no treatment effect on either component. The construction yields a single prespecified p-value together with interpretable component-wise effect estimates and confidence regions. We present both a parametric version and a semi-parametric version based on empirical likelihood, establish their large-sample calibration, and study their behavior in simulations. The methods are most advantageous when component effects can be obscured by the one-dimensional composite or when survivor-distribution features make standard combined-outcome tests inefficient. A clinical application to the IST-3 acute ischemic stroke trial illustrates the method using death by six months as the atom and EuroQol visual analogue score among survivors as the continuous component.

Keywords: Quality of Life, Randomized Controlled Trials,
Truncated Outcomes, Semi-Continuous Distributions, Empirical Likelihood

1 Introduction

Continuous outcomes such as quality-of-life scores, cognition scores, and days alive outside hospital are increasingly used in randomized trials of serious illness. When patients die before assessment, however, those outcomes are undefined rather than merely missing, creating the truncation-by-death problem (Frangakis and Rubin, 2002; Colantuoni et al., 2018). Composite outcomes that combine mortality with an otherwise continuous scale are common in critical care and related areas; familiar examples include ventilator-free days and days alive without life support (The COVID STEROID 2 Trial Group, 2021; Yehya et al., 2019).

One pragmatic response is to place death, or another substantively distinct event, at the bottom of a single ordered scale and analyze the resulting composite with a rank-based or related one-dimensional method. Worst-rank analyses and related composite rankings have been advocated precisely because they retain all randomized patients in the primary analysis (Lachin, 1999; McMahon and Harrell, 2000; Colantuoni et al., 2018). The same logic underlies the widespread use

of atom-at-zero outcomes such as ventilator-free days, where death is assigned the worst possible value on the scale (Yehya et al., 2019). Throughout, we refer to the Mann–Whitney–Wilcoxon rank-sum test simply as the Wilcoxon test (Wilcoxon, 1945; Mann and Whitney, 1947).

That strategy is often clinically sensible, but it targets the distribution of the composite outcome itself. It does not separate treatment effects on the probability of the atom from treatment effects on the continuous outcome among patients with observed values. When the two components move in different directions, or when treatment acts mainly through one of them, a one-dimensional analysis can be attenuated because opposite component effects partially cancel on the combined scale. The issue is therefore not only one of robustness or convenience; it is also one of estimand alignment.

Other approaches address truncated-by-death outcomes from different perspectives. Survivor-only analyses condition on a post-randomization event and therefore answer a conditional question that can differ materially from the intention-to-treat comparison. Principal-stratification and survivor-average-causal-effect approaches instead target causal effects in latent survivor strata, but they require assumptions that are not identified from the observed trial data alone (Frangakis and Rubin, 2002; Colantuoni et al., 2018). Composite-endpoint approaches avoid that latent-stratum target by working directly with an ordered endpoint, but then the estimand is tied to the chosen ranking (McMahon and Harrell, 2000; Colantuoni et al., 2018). Our goal in this paper is narrower and more operational: we target the joint observed-data null of no treatment effect on the atom probability and no treatment effect on the continuous component among those for whom it is observed.

Our construction combines ideas from two-part models for semi-continuous outcomes (Aitchison, 1955; Cragg, 1971) and empirical-likelihood inference for mean restrictions (Owen, 1988, 1991; Kitamura, 2001). We develop a factorized likelihood-ratio framework that yields a single prespecified p-value for the joint null together with interpretable component-wise effect estimates and confidence regions. We present both a parametric version, which is efficient under a credible model for the positive part, and a semi-parametric version, which relaxes that distributional commitment while retaining the same Bernoulli component. The resulting procedures preserve the clinically meaningful atom used in practice, but they test a sharper question than direct one-dimensional analyses of the combined outcome.

Although we motivate the method by quality of life truncated by death, the same structure arises whenever one special value has a substantively different meaning than the remainder of an otherwise continuous scale. The target is an observed-data treatment contrast, not a principal-stratum causal estimand. The rest of the paper is organized as follows. Section 2 introduces the statistical setup and the proposed tests. Section 3 reports a simulation study, Section 4 gives a clinical application, and Section 5 concludes. Additional simulation results are collected in Appendix A.

2 Method

Let data be independent and identically distributed random variables (Y_i, A_i, R_i, X_i) for $i = 1, \ldots, n$, where Y_i is a continuous outcome when it is defined, $A_i \in \{0, 1\}$ indicates whether the outcome is defined and observed, $R_i \in \{0, 1\}$ is the treatment indicator, and X_i is a p-dimensional vector of baseline covariates. Because Y_i is only meaningful when $A_i = 1$, we define the null

hypothesis of no treatment effect as the joint equality

$$\mathcal{L}(Y_i \mid A_i = 1, R_i = 1, X_i) = \mathcal{L}(Y_i \mid A_i = 1, R_i = 0, X_i),$$
$$\Pr(A_i = 1 \mid R_i = 1, X_i) = \Pr(A_i = 1 \mid R_i = 0, X_i).$$

In practice, this bivariate object is often replaced by a single combined outcome. Let $\mathcal{Y} \subset \mathbb{R}$ be a compact set containing the support of $Y_i \mid A_i = 1$, and let $\mathcal{E} \notin \mathcal{Y}$ be a distinguished symbol representing the undefined outcome. We define the outcome space by

$$\widetilde{\mathcal{Y}} = \mathcal{Y} \sqcup \{\mathcal{E}\},$$

where $\sqcup$ denotes disjoint union. We equip $\widetilde{\mathcal{Y}}$ with the σ-algebra

$$\widetilde{\mathcal{B}} = \{B,\ B \cup \{\mathcal{E}\} : B \in \mathcal{B}(\mathcal{Y})\},$$

where $\mathcal{B}(\mathcal{Y})$ is the Borel σ-algebra on $\mathcal{Y}$ induced by $\mathbb{R}$. Thus $\mathcal{E}$ is an abstract atom, distinct from every point of $\mathcal{Y}$, and $\widetilde{Y}_i$ is a $\widetilde{\mathcal{Y}}$-valued random variable. For fixed $e_0 \in \mathbb{R}$, we define the numerical representation map

$$c_{e_0} : \widetilde{\mathcal{Y}} \to \mathbb{R}, \qquad c_{e_0}(y) = y \text{ for } y \in \mathcal{Y}, \qquad c_{e_0}(\mathcal{E}) = e_0,$$

and arithmetic is performed on $c_{e_0}(\widetilde{Y}_i)$, not on the abstract atom itself. When no ambiguity arises, one may equivalently work directly on the numerical outcome space $\mathcal{Y} \cup \{e_0\} \subseteq \mathbb{R}$, thereby identifying the atom with the actual number e_0. Throughout most of the manuscript we use the atom-at-zero representation c_0, obtained by setting $e_0 = 0$.

$$\widetilde{Y}_i = \begin{cases} Y_i, & A_i = 1, \\ \mathcal{E}, & A_i = 0. \end{cases} \tag{1}$$

where $\mathcal{E}$ is a fixed atom representing the undefined outcome. The distribution of $\widetilde{Y}_i$ is therefore a mixture of a point mass at $\mathcal{E}$ and a continuous distribution on the support of $Y_i \mid A_i = 1$.

$$\widetilde{Y}_i^{\,0} := c_0(\widetilde{Y}_i) = \begin{cases} Y_i, & A_i = 1, \\ 0, & A_i = 0. \end{cases}$$

This is the real-valued combined outcome used whenever expectations or other arithmetic summaries are taken.

The conditional mean of the numerical combined outcome is

$$\mathrm{E}[\widetilde{Y}_i^{\,0} \mid R_i, X_i] = \mathrm{E}[Y_i \mid A_i = 1, R_i, X_i] \Pr(A_i = 1 \mid R_i, X_i).$$

This quantity can be identical between treatment groups even when treatment affects both the observation probability and the continuous component. To see this, let

$$\mathrm{E}[Y_i \mid A_i = 1, R_i = 0, X_i] = \mu(X_i), \qquad \mathrm{E}[Y_i \mid A_i = 1, R_i = 1, X_i] = \mu(X_i) + \mu_\delta(X_i),$$

Then

$$\begin{aligned} \Delta_{\widetilde{Y}}(X_i) &= \mathrm{E}[\widetilde{Y}_i^{\,0} \mid R_i = 1, X_i] - \mathrm{E}[\widetilde{Y}_i^{\,0} \mid R_i = 0, X_i] \\ &= \mu(X_i)\,[\Pr(A_i = 1 \mid R_i = 1, X_i) - \Pr(A_i = 1 \mid R_i = 0, X_i)] \end{aligned} \tag{2}$$

$$+ \mu_\delta(X_i) \Pr(A_i = 1 \mid R_i = 1, X_i).$$

Here $\Delta_{\widetilde{Y}}(X_i)$ is a contrast on the atom-at-zero numerical representation $\widetilde{Y}_i^0$, not a mean on the abstract disjoint-union space. Choosing

$$\mu_\delta(X_i) = \mu(X_i) \left[\Pr(A_i = 1 \mid R_i = 0, X_i) \Pr(A_i = 1 \mid R_i = 1, X_i)^{-1} - 1 \right]$$

forces $\Delta_{\widetilde{Y}}(X_i) = 0$ even though both components depend on treatment. Thus testing $\Delta_{\widetilde{Y}}(X_i) = 0$ is generally not equivalent to testing no treatment effect on the joint observed-data structure, and the latter requires two restrictions.

In the abstract formulation, $\mathcal{E}$ is a fixed prespecified atom representing the clinically worst state on the combined scale. When numerical summaries are required, we pass to a real-valued representation c_{e_0}; throughout most of the manuscript, the default choice is $e_0 = 0$. In applications where the observed endpoint truly has an atom at a real number such as 0, we may work directly on that numerical scale. The event $A_i = 0$ therefore denotes the substantive atom event, such as death before assessment or zero days alive out of hospital, rather than ordinary measurement missingness. Our estimands are defined with respect to the observed-data law of $(A_i, Y_i \mid A_i = 1)$, so identifiability is with respect to that observed-data structure rather than a latent principal stratum. The inferential development below assumes positive probabilities of $A_i = 1$ in both treatment groups, identifiable finite-dimensional models for the Bernoulli component, and, when the parametric procedure is used, a correctly specified model for the conditional outcome distribution.

To exploit that structure, we model the binary and continuous components separately. Let

$$\begin{aligned} A_i \mid R_i, X_i &\sim \text{Bernoulli}\left(\pi_i\right), \\ Y_i \mid A_i = 1, R_i, X_i &\sim F\left(\cdot; \mu_i, \Psi\right), \end{aligned}$$

where $\pi_i = \Pr(A_i = 1 \mid R_i, X_i)$ is the observation probability, $\mu_i = \mathrm{E}[Y_i \mid A_i = 1, R_i, X_i]$ is a mean parameter for the observed outcomes, and Ψ collects nuisance parameters governing the distribution of the continuous component. If $f(\cdot; \mu_i, \Psi)$ denotes the corresponding density, the observed-data likelihood is

$$L_n(\mu_i, \pi_i, \Psi) = \prod_{i=1}^{n} f\left(Y_i; \mu_i, \Psi\right)^{A_i} \pi_i^{A_i} \left(1 - \pi_i\right)^{1 - A_i}.$$

This factorizes as

$$L_n(\mu_i, \pi_i, \Psi) = L_{Y,n}(\mu_i, \Psi) L_{A,n}(\pi_i),$$

where

$$\begin{aligned} L_{Y,n}(\mu_i, \Psi) &= \prod_{i=1}^{n} f\left(Y_i; \mu_i, \Psi\right)^{A_i}, \\ L_{A,n}(\pi_i) &= \prod_{i=1}^{n} \pi_i^{A_i} \left(1 - \pi_i\right)^{1 - A_i}. \end{aligned}$$

The factorization follows from the likelihood construction; it does not require independence between the value of Y_i and the event $A_i = 1$.

We now impose finite-dimensional regression models on the two components. Let $s_\beta(x) = h_\beta(x)^\top \eta_\beta$ and $s_\mu(x) = h_\mu(x)^\top \eta_\mu$, where h_β and h_μ are prespecified transformations of the baseline covariates and η_β, η_μ are nuisance parameters. The working models used for covariate adjustment are

$$\operatorname{logit} \Pr(A_i = 1 \mid R_i, X_i) = \beta_0 + \beta_\delta R_i + s_\beta(X_i), \tag{3}$$

$$\mathrm{E}[Y_i \mid A_i = 1, R_i, X_i] = \mu_0 + \mu_\delta R_i + s_\mu(X_i), \tag{4}$$

so that treatment enters additively on the logit scale for the observation probability and additively on the mean scale for the continuous non-atom outcome. The parameter μ_δ is the conditional mean difference among observed outcomes at fixed covariates, and β_δ is the conditional log odds-ratio of being observed. We write $\alpha_\delta = \exp(\beta_\delta)$ for the corresponding odds ratio.

On the original outcome scale, μ_δ quantifies how treatment shifts the positive part of the outcome distribution among subjects with observed outcomes, whereas β_δ quantifies how treatment changes the odds that the outcome is observed at all. Reporting the pair therefore separates improvement in the continuous component from improvement in the probability of escaping the atom. This two-component contrast is the scientific target of the proposed likelihood-ratio test. It remains informative even when a single combined-outcome contrast such as $\Delta_{\widetilde{Y}}(X_i)$ is zero because the two components offset each other.

For fixed treatment contrasts $(\mu_\delta, \beta_\delta)$, the profile likelihood-ratio statistic is

$$\begin{aligned} W_n^P(\mu_\delta, \beta_\delta) &= W_{Y,n}^P(\mu_\delta) + W_{A,n}^P(\beta_\delta), \\ W_{Y,n}^P(\mu_\delta) &= -2 \log \frac{\sup\limits_{\mu_0, s_\mu, \Psi} L_{Y,n}(\mu_0, \mu_\delta, s_\mu, \Psi)}{\sup\limits_{\mu_0, \mu_\delta, s_\mu, \Psi} L_{Y,n}(\mu_0, \mu_\delta, s_\mu, \Psi)}, \\ W_{A,n}^P(\beta_\delta) &= -2 \log \frac{\sup\limits_{\beta_0, s_\beta} L_{A,n}(\beta_0, \beta_\delta, s_\beta)}{\sup\limits_{\beta_0, \beta_\delta, s_\beta} L_{A,n}(\beta_0, \beta_\delta, s_\beta)}. \end{aligned} \tag{5}$$

The null hypothesis of no treatment effect is tested with $W_n^P(0, 0)$.

2.1 Parametric likelihood-ratio test

In the parametric version, the continuous non-atom outcome distribution is specified fully. In the Gaussian working model considered here,

$$\begin{aligned} A_i \mid R_i = r, X_i = x &\sim \text{Bernoulli}\{\pi_r(x)\}, \\ Y_i \mid A_i = 1, R_i = r, X_i = x &\sim N\{\mu_r(x), \sigma^2\}, \end{aligned} \tag{6}$$

where

$$\pi_r(x) = \overset{-1}{\operatorname{logit}}\{\beta_0 + \beta_\delta r + s_\beta(x)\}, \qquad \mu_r(x) = \mu_0 + \mu_\delta r + s_\mu(x), \qquad r \in \{0, 1\}.$$

The residual variance σ^2 is common to the two treatment arms, conditional on the adjustment covariates. Let $\theta_P = (\beta_0, \beta_\delta, \eta_\beta^\top, \mu_0, \mu_\delta, \eta_\mu^\top, \sigma^2)^\top$ collect all finite-dimensional parameters. The observed-data likelihood under the Gaussian model is

$$L_n^P(\theta_P) = \prod_{i=1}^n \left[\pi_{R_i}(X_i)^{A_i} \{1 - \pi_{R_i}(X_i)\}^{1-A_i} \right]$$

$$\times \prod_{i=1}^{n} \left[(2\pi\sigma^2)^{-1/2} \exp\left\{ -\frac{\{Y_i - \mu_{R_i}(X_i)\}^2}{2\sigma^2} \right\} \right]^{A_i}. \tag{7}$$

Thus the likelihood is the product of a Bernoulli likelihood for whether the outcome escapes the atom and a Normal likelihood for the observed continuous values. The null model for the joint test imposes $\mu_\delta = 0$ and $\beta_\delta = 0$; the remaining regression parameters and σ^2 are nuisance parameters profiled out in Equation (5).

The regression formulation also induces marginal summaries standardized to a common covariate distribution. Let $\mathbb{P}_n$ denote the empirical distribution of the observed baseline covariates, so that $\mathbb{P}_n g(X) = n^{-1} \sum_{i=1}^{n} g(X_i)$. For $r \in \{0, 1\}$ define

$$\bar{\pi}_{r,n} = \mathbb{P}_n\{\pi_r(X)\}, \qquad \bar{\rho}_{r,n} = 1 - \bar{\pi}_{r,n}, \qquad \bar{\mu}_{r,n} = \mathbb{P}_n\{\mu_r(X)\}. \tag{8}$$

Here $\bar{\pi}_{r,n}$ is the observation probability that would be obtained if all subjects were assigned treatment arm r while retaining the empirical covariate distribution, $\bar{\rho}_{r,n}$ is the corresponding atom probability, and $\bar{\mu}_{r,n}$ is the standardized mean of the continuous outcome among non-atom observations. The component contrasts are

$$\delta_{\pi,n} = \bar{\pi}_{1,n} - \bar{\pi}_{0,n}, \qquad \delta_{\rho,n} = \bar{\rho}_{1,n} - \bar{\rho}_{0,n} = -\delta_{\pi,n}, \qquad \delta_{\mu,n} = \bar{\mu}_{1,n} - \bar{\mu}_{0,n}.$$

Under the additive mean model in Equation (4), $\delta_{\mu,n} = \mu_\delta$ for every empirical covariate distribution. By contrast, the observation-probability contrast $\delta_{\pi,n}$ is a marginal standardization of the conditional log-odds model, whereas β_δ and $\alpha_\delta = \exp(\beta_\delta)$ remain the conditional log odds-ratio and odds ratio, respectively.

For a real-valued representation c_{e_0} of the atom, the covariate-standardized combined-outcome contrast induced by the parametric model is

$$\Delta_n^{(e_0)} = \mathbb{P}_n\Big[e_0\{1 - \pi_1(X)\} + \pi_1(X)\mu_1(X) - e_0\{1 - \pi_0(X)\} - \pi_0(X)\mu_0(X) \Big]. \tag{9}$$

For the atom-at-zero representation used in most of the manuscript, Equation (9) reduces to $\mathbb{P}_n\{\pi_1(X)\mu_1(X) - \pi_0(X)\mu_0(X)\}$. These standardized quantities compare the two treatment arms at the same empirical covariate distribution, rather than comparing arm-specific fitted means over potentially different covariate distributions.

Proposition 1. *Assume that the parametric model for $Y_i \mid A_i = 1, R_i, X_i$ is correctly specified, that the regression models in Equations (3) and (4) are finite-dimensional and correctly specified, that the true parameter lies in the interior of the parameter space, that the model is identifiable with nonsingular Fisher information, and that $0 < \Pr(A_i = 1 \mid R_i = r, X_i) < 1$ almost surely for $r = 0, 1$. Let $n_r = \sum_{i=1}^{n} \mathbb{1}(R_i = r)$ and assume $n_0, n_1 \to \infty$ with $n_1/n_0 \to c \in (0, \infty)$. Then, under the null hypothesis $\mu_\delta = \beta_\delta = 0$, the profile likelihood-ratio statistic $W_n^P(0, 0)$ in Equation (5) converges in distribution to χ_2^2.*

Proof. Under the stated regularity conditions, the joint likelihood is a regular finite-dimensional likelihood with two scalar restrictions under the null: one in the continuous component and one in the Bernoulli component. Because the likelihood factorizes into $L_{Y,n}$ and $L_{A,n}$, the joint likelihood-ratio statistic is exactly the sum of the corresponding component likelihood-ratio statistics. Wilks' theorem (Wilks, 1938) therefore yields the χ_2^2 limit. □

The same fitted conditional laws determine both the component likelihood-ratio statistic and the standardized summaries in Equations (8)–(9). Thus covariate adjustment has two distinct roles: it defines the conditional restrictions tested by the likelihood ratio, and it yields marginal summaries by averaging the fitted arm-specific conditional quantities over a common empirical distribution of X_i.

2.2 Semi-parametric likelihood-ratio test

The parametric test depends on the chosen distributional model for the observed outcomes. To relax that assumption, we replace the Gaussian likelihood for the continuous non-atom outcome by an empirical likelihood for the conditional mean restrictions, while retaining the parametric Bernoulli likelihood for A_i. The observation component is therefore still governed by Equation (3), but no Normality, homoscedasticity, or parametric shape assumption is imposed on the distribution of $Y_i \mid A_i = 1, R_i, X_i$ beyond the finite-dimensional conditional mean model in Equation (4).

Let $\mathcal{J} = \{i : A_i = 1\}$ and $M_n = |\mathcal{J}|$. For $i \in \mathcal{J}$ define the observed-outcome design vector

$$Z_i = \{1, R_i, h_\mu(X_i)^\top\}^\top, \qquad \theta = (\mu_0, \mu_\delta, \eta_\mu^\top)^\top,$$

so that $\mu_{R_i}(X_i) = Z_i^\top \theta$. The semi-parametric continuous component is characterized by the moment restriction

$$\mathrm{E}\Big[Z_i\{Y_i - Z_i^\top \theta\} \mid A_i = 1\Big] = 0. \tag{10}$$

For a fixed candidate value m_δ of the treatment mean coefficient, write $\theta(m_\delta, \nu)$ for the vector obtained by setting the treatment component of θ equal to m_δ and collecting the remaining coefficients in the nuisance vector ν. Define

$$g_i\{m_\delta, \nu\} = Z_i \left[Y_i - Z_i^\top \theta(m_\delta, \nu)\right], \qquad i \in \mathcal{J}.$$

The profile empirical likelihood ratio for the continuous component is

$$R^{\mathrm{E}}_{Y,n}(m_\delta) = \sup_{\nu, \{p_i : i \in \mathcal{J}\}} \left\{ \prod_{i \in \mathcal{J}} M_n p_i \right\}, \tag{11}$$

where the supremum is over weights satisfying

$$p_i \geq 0, \qquad \sum_{i \in \mathcal{J}} p_i = 1, \qquad \sum_{i \in \mathcal{J}} p_i g_i\{m_\delta, \nu\} = 0.$$

The corresponding empirical-likelihood deviance is

$$\widetilde{W}^P_{Y,n}(m_\delta) = -2 \log R^{\mathrm{E}}_{Y,n}(m_\delta). \tag{12}$$

Equivalently, for each feasible (m_δ, ν) there is a Lagrange multiplier $\lambda(m_\delta, \nu)$ of the same dimension as Z_i such that

$$\sum_{i \in \mathcal{J}} \frac{g_i\{m_\delta, \nu\}}{1 + \lambda(m_\delta, \nu)^\top g_i\{m_\delta, \nu\}} = 0, \qquad 1 + \lambda(m_\delta, \nu)^\top g_i\{m_\delta, \nu\} > 0 \quad \text{for all } i \in \mathcal{J}.$$

In dual form,

$$\widetilde{W}^P_{Y,n}(m_\delta) = \inf_\nu 2 \sum_{i \in \mathcal{J}} \log \left[1 + \lambda(m_\delta, \nu)^\top g_i\{m_\delta, \nu\}\right], \tag{13}$$

with the value set to infinity when the empirical-likelihood constraints are infeasible. The fitted non-atom mean functions are standardized in the same way as in Equation (8):

$$\bar{\mu}^{\mathrm{SP}}_{r,n}(\theta) = \mathbb{P}_n\{(1, r, h_\mu(X)^\top)\theta\}, \qquad r \in \{0, 1\}.$$

Under the additive mean model, $\bar{\mu}^{\mathrm{SP}}_{1,n}(\theta) - \bar{\mu}^{\mathrm{SP}}_{0,n}(\theta) = \mu_\delta$. The semi-parametric standardized combined-outcome contrast is obtained by inserting this empirical-likelihood mean model and the fitted logistic observation probabilities into Equation (9). Thus the continuous distribution itself is left unspecified, but the mean component entering the standardized contrast is defined by the estimating equations in Equation (10).

When X_i is absent, the empirical-likelihood regression construction reduces to the familiar two-sample empirical likelihood for the observed outcome means. Let

$$\mathcal{I}_0 = \{i = 1, \ldots, n : R_i = 0, A_i = 1\}, \qquad \mathcal{I}_1 = \{i = 1, \ldots, n : R_i = 1, A_i = 1\}.$$

In the no-covariate case, the standard empirical-likelihood dual representation gives, for fixed (μ_0, μ_δ), the empirical-likelihood deviance

$$\widetilde{W}_{Y,n}(\mu_0, \mu_\delta) = 2\sum_{i=1}^{n}\left[(1 - R_i)\log\left(1 + \lambda_0(\widetilde{Y}_i^{\,0} - A_i\mu_0)\right) + R_i \log\left(1 + \lambda_1(\widetilde{Y}_i^{\,0} - A_i(\mu_0 + \mu_\delta))\right)\right], \tag{14}$$

where terms with $A_i = 0$ contribute zero because $\widetilde{Y}_i^{\,0} = 0$ in that case. Writing

$$n_0 = \sum_{i=1}^{n}(1 - R_i)A_i = |\mathcal{I}_0|, \qquad n_1 = \sum_{i=1}^{n} R_i A_i = |\mathcal{I}_1|,$$

the dual parameters λ_0 and λ_1 solve

$$\frac{1}{n_0}\sum_{i=1}^{n}(1 - R_i)\frac{\widetilde{Y}_i^{\,0} - A_i\mu_0}{1 + \lambda_0(\widetilde{Y}_i^{\,0} - A_i\mu_0)} = 0,$$
$$\frac{1}{n_1}\sum_{i=1}^{n} R_i\frac{\widetilde{Y}_i^{\,0} - A_i(\mu_0 + \mu_\delta)}{1 + \lambda_1(\widetilde{Y}_i^{\,0} - A_i(\mu_0 + \mu_\delta))} = 0.$$

The profile version is obtained by minimizing over the nuisance mean:

$$\widetilde{W}^P_{Y,n}(\mu_\delta) = \inf_{\mu_0 \in \mathbb{R}} \widetilde{W}_{Y,n}(\mu_0, \mu_\delta).$$

The admissible values of μ_0 are those for which the corresponding constrained means lie in the convex hull of the observed outcomes in each group. For interior feasible means, the dual parameters λ_0 and λ_1 are determined by one-dimensional root equations with unique roots on their admissible intervals, and the profiled nuisance mean is then minimized over its feasible set. If a candidate constraint falls outside the convex-hull region, the corresponding empirical-likelihood statistic is treated as infinite.

The semi-parametric test statistic is then

$$\widetilde{W}^P_n(\mu_\delta, \beta_\delta) = \widetilde{W}^P_{Y,n}(\mu_\delta) + W^P_{A,n}(\beta_\delta). \tag{15}$$

The first term is the profile empirical-likelihood deviance in Equation (12), or equivalently Equation (14) in the no-covariate special case; the second term is the Bernoulli profile likelihood-ratio statistic from Equation (5).

Proposition 2 (Covariate-adjusted semi-parametric likelihood-ratio test)**.** *Consider the null hypothesis*

$$H_0: \ \mu_\delta = 0, \quad \beta_\delta = 0,$$

equivalently, under the additive adjustment models, no covariate-standardized difference in the non-atom mean and no covariate-specific log odds-ratio for observation. Assume that (Y_i, A_i, R_i, X_i) *are independent observations from a common law; that for some* $\varepsilon > 0$, $\varepsilon \le \Pr(R_i = 1 \mid X_i) \le 1 - \varepsilon$ *and* $\varepsilon \le \Pr(A_i = 1 \mid R_i = r, X_i) \le 1 - \varepsilon$ *almost surely for* $r = 0, 1$*; that the logistic model in Equation (3) and the conditional mean model in Equation (4) are correctly specified; that the true finite-dimensional parameters lie in the interior of their parameter spaces; that the Fisher information for the Bernoulli component is nonsingular; and that, for the empirical-likelihood component,* $\mathrm{E}\,\|Z_i\{Y_i - Z_i^\top \theta_0\}\|^{2+\eta} < \infty$ *for some* $\eta > 0$*, the covariance matrix of* $Z_i\{Y_i - Z_i^\top \theta_0\}$ *conditional on* $A_i = 1$ *is nonsingular, and the derivative matrix of the moment condition in Equation (10) has full column rank. Then, under* H_0*,*

$$\widetilde{W}_n^P(0,0) = \widetilde{W}_{Y,n}^P(0) + W_{A,n}^P(0) \ \xrightarrow{d} \ \chi_2^2.$$

The two degrees of freedom correspond to the two scalar restrictions tested jointly: $\mu_\delta = 0$ *in the continuous mean component and* $\beta_\delta = 0$ *in the observation component.*

Proof. The Bernoulli component is a regular finite-dimensional likelihood with one scalar restriction, so Wilks' theorem gives $W_{A,n}^P(0) \xrightarrow{d} \chi_1^2$. Conditional on the non-atom sample, the continuous component is an empirical likelihood for estimating equations with a single profiled scalar restriction. The empirical-likelihood Wilks theorem for estimating equations therefore gives $\widetilde{W}_{Y,n}^P(0) \xrightarrow{d} \chi_1^2$ under the stated moment, rank, and interiority conditions (Owen, 1991; Qin and Lawless, 1994). The Bernoulli score is a function of (A_i, R_i, X_i), whereas the empirical-likelihood estimating function has conditional mean zero given $(A_i = 1, R_i, X_i)$; hence the two components are asymptotically block-orthogonal. The sum in Equation (15) therefore has the χ_2^2 null limit. □

The adjusted parametric and semi-parametric analyses consequently target the same two-component treatment structure. Both use the logistic model for the probability of escaping the atom and both summarize treatment effects at the empirical covariate distribution through Equations (8)–(9). They differ in the continuous non-atom component: the parametric analysis uses a fully specified Gaussian conditional distribution with common residual variance, whereas the semi-parametric analysis uses only the regression moment restrictions in Equation (10) and leaves the remaining shape of the continuous distribution unspecified. The semi-parametric likelihood-ratio test is therefore less sensitive to non-Normality of the continuous outcomes, at the cost of relying more directly on large-sample empirical-likelihood approximations and on feasibility of the moment constraints.

2.3 Inference and interpretation

Confidence regions follow by inverting the corresponding likelihood-ratio statistic. For the parametric procedure, an asymptotic $(1-\alpha)100\%$ simultaneous confidence region for $(\mu_\delta, \beta_\delta)$ is

$$\mathrm{CI}_{1-\alpha}^{(\mu_\delta,\beta_\delta)} = \left\{(m,b) \in \mathbb{R}^2 : W_n^P(m,b) \le \chi_{2,1-\alpha}^2\right\}.$$

For the semi-parametric procedure, one replaces W_n^P by $\widetilde{W}_n^P$. Correspondingly, the marginal intervals are

$$\mathrm{CI}_{1-\alpha}^{\mu_\delta} = \left\{ m \in \mathbb{R} : W_{Y,n}^P(m) \le \chi^2_{1,1-\alpha} \right\} \quad \text{or} \quad \left\{ m \in \mathbb{R} : \widetilde{W}_{Y,n}^P(m) \le \chi^2_{1,1-\alpha} \right\},$$
$$\mathrm{CI}_{1-\alpha}^{\beta_\delta} = \left\{ b \in \mathbb{R} : W_{A,n}^P(b) \le \chi^2_{1,1-\alpha} \right\}.$$

Here $\chi^2_{q,1-\alpha}$ denotes the $(1-\alpha)$ quantile of a chi-square distribution with q degrees of freedom. We recommend reporting the simultaneous confidence region because it displays the joint uncertainty about the effect on the continuous and binary components.

Many applied reports also require an interval on the numerical combined-outcome scale. Such an interval concerns a derived scalar functional, denoted here by Δ or $\Delta^{(e_0)}$, rather than the two-component parameter itself. It therefore answers a different inferential question from the region for $(\mu_\delta, \beta_\delta)$: it summarizes the consequence of the atom and continuous components after the atom has been assigned the numerical value e_0 through the representation map c_{e_0}. This scalar summary is useful for reporting effects on the original numerical scale, but it is secondary to the two-component analysis because it necessarily collapses changes in the observation probability and changes in the non-atom mean into a single number.

To define the likelihood-based intervals uniformly, let

$$\mathcal{W}_n(m,b) = \begin{cases} W_n^P(m,b), & \text{for the parametric likelihood,} \\ \widetilde{W}_n^P(m,b), & \text{for the semi-parametric likelihood.} \end{cases} \tag{16}$$

For fixed component contrasts (m,b), let $\widehat{\pi}_r(x;b)$ denote the fitted observation probability in arm r obtained after profiling the Bernoulli nuisance parameters under the restriction $\beta_\delta = b$. Similarly, let $\widehat{\mu}_r(x;m)$ denote the fitted conditional non-atom mean in arm r obtained after profiling the continuous-component nuisance parameters under the restriction $\mu_\delta = m$. In the parametric analysis this profiling is with respect to the Gaussian likelihood; in the semi-parametric analysis it is with respect to the empirical-likelihood estimating-equation constraints. The profiled combined-outcome contrast induced by (m,b) and the atom representation c_{e_0} is

$$\begin{aligned} g_n^{(e_0)}(m,b) = \mathbb{P}_n\Big[& e_0\{1-\widehat{\pi}_1(X;b)\} + \widehat{\pi}_1(X;b)\widehat{\mu}_1(X;m) \\ & - e_0\{1-\widehat{\pi}_0(X;b)\} - \widehat{\pi}_0(X;b)\widehat{\mu}_0(X;m)\Big]. \end{aligned} \tag{17}$$

When there are no adjustment covariates and $e_0 = 0$, this reduces to

$$g_n^{(0)}(m,b) = \widehat{\pi}_1(b)\widehat{\mu}_1(m) - \widehat{\pi}_0(b)\widehat{\mu}_0(m),$$

where $\widehat{\pi}_r(b)$ and $\widehat{\mu}_r(m)$ are the corresponding arm-specific profiled probabilities and means. The fitted likelihood-based estimate of the numerical combined-outcome contrast is $g_n^{(e_0)}(\widehat{\mu}_\delta, \widehat{\beta}_\delta)$, where $(\widehat{\mu}_\delta, \widehat{\beta}_\delta)$ denotes the unrestricted component estimate.

In the same no-covariate setting, the profiled Bernoulli probabilities can be written explicitly. For fixed b, let

$$\widehat{\beta}_0(b) \in \arg\min_{\beta_0 \in \mathbb{R}} \sum_{i=1}^n \left[\log\{1+\exp(\beta_0 + bR_i)\} - A_i(\beta_0 + bR_i) \right],$$

and set

$$\widehat{\pi}_r(b) = \mathrm{logit}^{-1}\{\widehat{\beta}_0(b) + rb\}, \qquad r \in \{0,1\}.$$

For fixed m, the profiled non-atom means are

$$\widehat{\mu}_0(m) \in \begin{cases} \arg\max_{\mu_0,\Psi} L_{Y,n}(\mu_0, m, \Psi), & \text{parametric}, \\ \arg\min_{\mu_0\in\mathbb{R}} \widetilde{W}_{Y,n}(\mu_0, m), & \text{semi-parametric}, \end{cases} \qquad \widehat{\mu}_1(m) = \widehat{\mu}_0(m) + m.$$

These are the quantities entering the no-covariate map $g_n^{(0)}(m, b)$.

The first interval is the direct Welch interval for the atom-at-zero numerical outcome. Let $n_r = \sum_{i=1}^n \mathbb{1}(R_i = r)$ and define

$$\widehat{\Delta}_{\mathrm{W}} = \bar{\widetilde{Y}}_1^{\,0} - \bar{\widetilde{Y}}_0^{\,0}, \qquad \bar{\widetilde{Y}}_r^{\,0} = n_r^{-1} \sum_{i:R_i=r} \widetilde{Y}_i^{\,0},$$

with arm-specific sample variances

$$S_r^2 = (n_r - 1)^{-1} \sum_{i:R_i=r} \left(\widetilde{Y}_i^{\,0} - \bar{\widetilde{Y}}_r^{\,0}\right)^2, \qquad r \in \{0, 1\}.$$

The Welch standard error and Satterthwaite degrees of freedom are

$$\widehat{\mathrm{se}}_{\mathrm{W}}^2 = \frac{S_1^2}{n_1} + \frac{S_0^2}{n_0}, \qquad \widehat{\nu} = \frac{\left(S_1^2/n_1 + S_0^2/n_0\right)^2}{\frac{(S_1^2/n_1)^2}{n_1-1} + \frac{(S_0^2/n_0)^2}{n_0-1}},$$

with the usual limiting interpretation if one of the variance terms is zero. The resulting interval is

$$\mathrm{CI}_{1-\alpha}^{\Delta,\mathrm{W}} = \left[\widehat{\Delta}_{\mathrm{W}} - t_{\widehat{\nu},1-\alpha/2}\widehat{\mathrm{se}}_{\mathrm{W}}, \widehat{\Delta}_{\mathrm{W}} + t_{\widehat{\nu},1-\alpha/2}\widehat{\mathrm{se}}_{\mathrm{W}}\right], \tag{18}$$

where $t_{\widehat{\nu},1-\alpha/2}$ is the corresponding Student quantile. This interval treats $\widetilde{Y}^{\,0}$ as an ordinary real-valued outcome and therefore does not use the two-component likelihood structure.

The second interval is obtained by projecting the simultaneous likelihood-ratio confidence region for the component parameter onto the $\Delta^{(e_0)}$ scale. Define

$$\mathcal{C}_{2,n}(1-\alpha) = \left\{(m, b) \in \mathbb{R}^2 : \mathcal{W}_n(m, b) \le \chi^2_{2,1-\alpha}\right\}. \tag{19}$$

The projected likelihood-ratio interval is the image of this region under the profiled map in Equation (17):

$$\mathrm{CI}_{1-\alpha}^{\Delta,\mathrm{proj}} = \left[\inf_{(m,b)\in\mathcal{C}_{2,n}(1-\alpha)} g_n^{(e_0)}(m, b), \sup_{(m,b)\in\mathcal{C}_{2,n}(1-\alpha)} g_n^{(e_0)}(m, b)\right]. \tag{20}$$

Equivalently, its endpoints are the constrained optimization problems

$$L_{\mathrm{proj}} = \inf_{m,b} g_n^{(e_0)}(m, b) \quad \text{subject to} \quad \mathcal{W}_n(m, b) \le \chi^2_{2,1-\alpha}, \tag{21}$$

$$U_{\mathrm{proj}} = \sup_{m,b} g_n^{(e_0)}(m, b) \quad \text{subject to} \quad \mathcal{W}_n(m, b) \le \chi^2_{2,1-\alpha}. \tag{22}$$

Because this construction preserves the entire two-dimensional confidence set before mapping it to a scalar, it inherits simultaneous coverage for $(\mu_\delta, \beta_\delta)$ and is typically conservative for $\Delta^{(e_0)}$ alone.

The third interval profiles the likelihood ratio directly over the scalar constraint $\Delta^{(e_0)} = d$. For each $d \in \mathbb{R}$, define

$$T_n(d) = \inf_{m,b} \left\{ \mathcal{W}_n(m,b) : g_n^{(e_0)}(m,b) = d \right\}. \tag{23}$$

Thus evaluating $T_n(d)$ is itself a constrained optimization problem: minimize the joint likelihood-ratio surface over component contrasts whose profiled numerical combined-outcome contrast is exactly d. The profile likelihood-ratio interval is

$$\mathrm{CI}_{1-\alpha}^{\Delta,\mathrm{prof}} = \left\{ d \in \mathbb{R} : T_n(d) \le \chi^2_{1,1-\alpha} \right\}, \tag{24}$$

with endpoints

$$L_{\mathrm{prof}} = \inf\{d : T_n(d) \le \chi^2_{1,1-\alpha}\}, \qquad U_{\mathrm{prof}} = \sup\{d : T_n(d) \le \chi^2_{1,1-\alpha}\}.$$

The one-degree-of-freedom calibration reflects that the null constraint is on the single scalar functional $g_n^{(e_0)}(m,b)$, while the remaining component direction is profiled out.

When the infimum defining $T_n(d)$ is attained on each non-empty fiber $\{(m,b) : g_n^{(e_0)}(m,b) = d\}$, the accepted Δ values can equivalently be written as the image of the one-degree-of-freedom likelihood-ratio sublevel set,

$$\left\{ g_n^{(e_0)}(m,b) : \mathcal{W}_n(m,b) \le \chi^2_{1,1-\alpha} \right\}.$$

This representation is useful for numerical evaluation and motivates the finite-grid approximation described in Appendix B.

In the unadjusted Gaussian model, the profile map has a closed form in the continuous component. Let

$$N_+ = \sum_{i=1}^n A_i, \qquad \overline{Y}_+ = \frac{1}{N_+} \sum_{i:A_i=1} Y_i, \qquad \overline{R}_+ = \frac{1}{N_+} \sum_{i:A_i=1} R_i.$$

For fixed m, profiling the Gaussian likelihood over the common intercept gives

$$\widehat{\mu}_0(m) = \overline{Y}_+ - \overline{R}_+ m, \qquad \widehat{\mu}_1(m) = \overline{Y}_+ + (1 - \overline{R}_+) m.$$

Substitution into $g_n^{(0)}$ yields the affine representation

$$\begin{aligned} g_n^{(0)}(m,b) = {} & \{\widehat{\pi}_1(b) - \widehat{\pi}_0(b)\}\overline{Y}_+ \\ & + \{\widehat{\pi}_1(b)(1 - \overline{R}_+) + \widehat{\pi}_0(b)\overline{R}_+\} m. \end{aligned} \tag{25}$$

Thus, for fixed d and b, the compatible mean contrast is

$$m^\star(d,b) = \frac{d - \{\widehat{\pi}_1(b) - \widehat{\pi}_0(b)\}\overline{Y}_+}{\widehat{\pi}_1(b)(1 - \overline{R}_+) + \widehat{\pi}_0(b)\overline{R}_+},$$

whenever the denominator is positive, and the profile statistic reduces to

$$T_n(d) = \inf_{b \in \mathbb{R}} \mathcal{W}_n\{m^\star(d,b), b\}.$$

For the semi-parametric likelihood, the map $m \mapsto g_n^{(0)}(m,b)$ is generally not affine. The corresponding reduction is obtained by profiling over the feasible compatible mean contrasts:

$$T_n(d) = \inf_{b \in \mathbb{R}} \inf \left\{ \mathcal{W}_n(m,b) : g_n^{(0)}(m,b) = d \right\},$$

with the inner infimum treated as infinite when no feasible empirical-likelihood mean contrast satisfies the scalar constraint.

Numerical endpoint calculation. The projected interval endpoints in Equation (20) may be computed as constrained optimization problems over a rectangular search box $B_n \subset \mathbb{R}^2$ on the (m, b) scale. Let $c_{2,\alpha} = \chi^2_{2,1-\alpha}$. The finite-box endpoint problems are

$$L^{\text{proj}}_{1-\alpha}(B_n) = \inf_{(m,b)\in B_n} \left\{ g_n^{(e_0)}(m, b) : \mathcal{W}_n(m, b) \le c_{2,\alpha} \right\}, \tag{26}$$

$$U^{\text{proj}}_{1-\alpha}(B_n) = \sup_{(m,b)\in B_n} \left\{ g_n^{(e_0)}(m, b) : \mathcal{W}_n(m, b) \le c_{2,\alpha} \right\}. \tag{27}$$

A practical unconstrained objective is obtained by replacing the hard likelihood-ratio constraint with a one-sided quadratic penalty. With $x_+ = \max(x, 0)$ and a large data-scaled penalty constant $\kappa_n > 0$, the lower- and upper-endpoint objectives are

$$Q^{\text{proj},L}_{n,\alpha}(m, b) = g_n^{(e_0)}(m, b) + \kappa_n \{\mathcal{W}_n(m, b) - c_{2,\alpha}\}^2_+, \tag{28}$$

$$Q^{\text{proj},U}_{n,\alpha}(m, b) = -g_n^{(e_0)}(m, b) + \kappa_n \{\mathcal{W}_n(m, b) - c_{2,\alpha}\}^2_+. \tag{29}$$

After numerical minimization of Equation (28) or Equation (29) within B_n, the candidate is projected back to the feasible boundary if necessary. Specifically, let $(\widehat{m}, \widehat{b})$ denote the unrestricted estimate and let $(m^\dagger, b^\dagger)$ denote the optimizer proposal. A one-dimensional binary search is performed on the line segment joining these two points to find the last point satisfying $\mathcal{W}_n(m, b) \le c_{2,\alpha}$. If an endpoint remains close to the boundary of B_n, the search box is expanded and the optimization is repeated.

For the profile interval, the statistic $T_n(d)$ is evaluated for candidate values d of the combined-outcome contrast. In the unadjusted Gaussian case, this uses the one-dimensional profile over b displayed above. In the semi-parametric case, the compatible mean contrast is obtained numerically. For fixed (d, b), the empirical-likelihood feasible range for m is

$$\left[\min_{j\in\mathcal{I}_1} Y_j - \max_{i\in\mathcal{I}_0} Y_i,\ \max_{j\in\mathcal{I}_1} Y_j - \min_{i\in\mathcal{I}_0} Y_i \right].$$

If the function $m \mapsto g_n^{(0)}(m, b) - d$ changes sign on this interval, the compatible value is found by one-dimensional root finding. Otherwise, the calculation minimizes the squared discrepancy

$$M_n(m; d, b) = \{g_n^{(0)}(m, b) - d\}^2$$

over the feasible interval and accepts the resulting candidate only if the remaining discrepancy is below numerical tolerance; if not, the profile value for that b is treated as infeasible. Once the function $d \mapsto T_n(d)$ can be evaluated, the lower and upper profile endpoints are obtained as the two roots of

$$T_n(d) = \chi^2_{1,1-\alpha}$$

on either side of the fitted value $g_n^{(e_0)}(\widehat{\mu}_\delta, \widehat{\beta}_\delta)$. The roots are bracketed by step-doubling away from the fitted value and then refined by bisection until the requested numerical tolerance is reached.

Proposition 3 (Likelihood-ratio intervals for the combined-outcome contrast)**.** *Let $\theta_\delta = (\mu_\delta, \beta_\delta)^\top$. Suppose that $\mathcal{W}_n$ satisfies the regularity conditions for the relevant parametric or semi-parametric likelihood-ratio theory in a neighborhood of the true value $\theta_{\delta 0}$. Assume further that $g_n^{(e_0)}$ converges uniformly in probability on this neighborhood to a continuously differentiable population map $g^{(e_0)}$, that $\nabla g^{(e_0)}(\theta_{\delta 0}) \neq 0$, and that $\Delta_0 = g^{(e_0)}(\theta_{\delta 0})$. Then*

$$T_n(\Delta_0) \xrightarrow{d} \chi^2_1,$$

and the profile interval in Equation (24) has asymptotic coverage $1-\alpha$ *for* Δ_0*. Moreover, if* $\mathcal{C}_{2,n}(1-\alpha)$ *is an asymptotic* $(1-\alpha)100\%$ *confidence region for* $\theta_{\delta 0}$*, then the projected interval in Equation (20) has at least asymptotic coverage* $1-\alpha$ *for* Δ_0*.*

Proof. The likelihood-ratio surface is locally quadratic under the regularity conditions for the corresponding parametric likelihood or empirical-likelihood estimating-equation model. Since $g^{(e_0)}$ is a smooth scalar map with nonzero gradient at $\theta_{\delta 0}$, the null hypothesis $g^{(e_0)}(\theta_\delta) = \Delta_0$ imposes one regular scalar restriction on the two-component parameter. Wilks' theorem and the empirical-likelihood analogue therefore give the χ^2_1 limit for the profiled statistic. For the projected interval, the event $\theta_{\delta 0} \in \mathcal{C}_{2,n}(1-\alpha)$ implies, by construction and continuity of $g_n^{(e_0)}$, that $g_n^{(e_0)}(\theta_{\delta 0})$ lies in the image interval up to an asymptotically negligible error. The stated coverage follows from the coverage of the two-dimensional confidence region. □

The three intervals therefore describe related but distinct targets. The Welch interval in Equation (18) is a direct one-dimensional interval for the mean difference of the coded numerical outcome $\widetilde{Y}^0$; it is simple and relatively model-light, but it does not distinguish whether the contrast arises from the atom probability, the continuous non-atom component, or both. The projected interval in Equation (20) is a likelihood-based summary derived from the simultaneous two-component confidence region; it is appropriate when $\Delta^{(e_0)}$ is reported as a secondary scalar summary while maintaining coherence with the joint analysis, and it is usually wider because it retains simultaneous uncertainty in both μ_δ and β_δ. The profile interval in Equation (24) is instead a direct likelihood-ratio interval for the scalar functional $\Delta^{(e_0)}$; it profiles over nuisance component directions and is therefore more focused on the numerical combined-outcome contrast. Thus the profile and projected intervals remain tied to the two-part likelihood decomposition, whereas the Welch interval is an analysis of the numerical representation alone.

In practice, the likelihood-based Δ intervals may also be approximated by evaluating the joint likelihood-ratio surface and the profiled Δ map on a finite grid of component contrasts. Appendix B gives the corresponding grid construction.

2.4 Bayesian two-group comparison

The preceding likelihood-based development admits a natural Bayesian counterpart. We again suppress X_i and work in the no-covariate setting, because this is the clearest setting in which to formulate the two-part observed-data law itself. Let $\widetilde{Y}_i^{(e_0)} = c_{e_0}(\widetilde{Y}_i)$ denote the numerical representation of the combined outcome for a fixed atom value $e_0 \in \mathbb{R}$. In the two-group comparison without covariates, the primary objects are the arm-specific observed-data laws. It is therefore more natural to fit treatment-stratified models than to impose a common mixture across treatment arms. A common mixture with only arm-specific shifts or scales would encode a priori similarity in distributional shape across arms, whereas in the present setting treatment may affect both the atom probability and the shape of the continuous non-atom distribution. Separate arm-specific mixtures align directly with the scientific question of whether the full observed-data law differs between treatment groups.

Let $\mathcal{C} \subseteq \mathbb{R}$ denote the support of the continuous non-atom outcomes on the chosen numerical scale. For continuous-support models, the examples below use $\mathcal{C} = \mathbb{R}$ or $\mathcal{C} = (0, \infty)$; bounded reported-score models replace the continuous conditional density by a probability mass function on a finite

score set. Let $\lambda_{\mathcal{C}}$ denote Lebesgue measure on $\mathcal{C}$, and define the dominating measure

$$\nu_{e_0} = \delta_{e_0} + \lambda_{\mathcal{C}}.$$

For each treatment arm $r \in \{0, 1\}$, we write the law of the numerical combined outcome as

$$\mathcal{L}(\widetilde{Y}_i^{(e_0)} \mid R_i = r) = \rho_r \delta_{e_0} + \pi_r F_r^c, \qquad \pi_r = 1 - \rho_r, \qquad F_r^c \ll \lambda_{\mathcal{C}}, \tag{30}$$

where $\rho_r = \Pr(\widetilde{Y}_i^{(e_0)} = e_0 \mid R_i = r)$ is the arm-specific atom probability and F_r^c is an absolutely continuous probability measure on $\mathcal{C}$ describing the non-atom outcomes. Equivalently, with f_r^c denoting the density of F_r^c with respect to $\lambda_{\mathcal{C}}$, the density of $\widetilde{Y}_i^{(e_0)} \mid R_i = r$ with respect to ν_{e_0} is

$$q_r(y) = \rho_r \mathbb{1}\{y = e_0\} + \pi_r f_r^c(y) \mathbb{1}\{y \in \mathcal{C}\}.$$

One could instead start directly from a single density with respect to the extended measure ν_{e_0} on $\mathcal{C} \cup \{e_0\}$. For the present problem, however, the explicit singular-plus-absolutely-continuous decomposition in Equation (30) is clearer, because it preserves the atom probability and the non-atom distribution as separate scientific objects and mirrors the observed-data factorization used in the likelihood-based analysis.

The main modeling task is therefore to specify a flexible prior for F_r^c. This is precisely the setting in which a Dirichlet process mixture model is attractive. The conditional non-atom distribution may be skewed, heavy-tailed, or multimodal, and a single parametric family can be too restrictive even when the scientific summaries of interest remain simple. By modeling F_r^c through a nonparametric mixture, one can accommodate complex distributional shape while retaining interpretable treatment-arm summaries such as the atom probability, the mean among non-atom outcomes, and contrasts derived from them.

Let $(\Theta, \mathcal{T})$ denote the parameter space for a support-matched kernel $K(\cdot \mid \theta)$ on $\mathcal{C}$. An arm-specific mixing distribution G_r is said to follow a Dirichlet process with concentration parameter $\alpha_r > 0$ and base measure G_0, written $G_r \sim \mathrm{DP}(\alpha_r, G_0)$, if for every measurable partition $(B_1, \ldots, B_m)$ of Θ,

$$(G_r(B_1), \ldots, G_r(B_m)) \sim \mathrm{Dirichlet}(\alpha_r G_0(B_1), \ldots, \alpha_r G_0(B_m))$$

(Ferguson, 1973). When G_r is used as a mixing distribution,

$$F_r^c(B) = \int_\Theta K(B \mid \theta)\, G_r(d\theta), \qquad B \subseteq \mathcal{C},$$

which yields the familiar Dirichlet process mixture model (Antoniak, 1974). By Sethuraman's stick-breaking representation (Sethuraman, 1994),

$$G_r = \sum_{h=1}^{\infty} w_{rh} \delta_{\theta_{rh}}, \qquad w_{r1} = v_{r1}, \qquad w_{rh} = v_{rh} \prod_{\ell=1}^{h-1} (1 - v_{r\ell}), \quad h \geq 2,$$

with

$$v_{rh} \mid \alpha_r \sim \mathrm{Beta}(1, \alpha_r), \qquad \theta_{rh} \overset{\mathrm{iid}}{\sim} G_0.$$

For computation we use the standard finite truncated approximation (Ishwaran and James, 2001), retaining H mixture components and assigning the remaining stick mass to the last retained component. This approximation gives a finite-dimensional posterior computation while retaining the stick-breaking interpretation.

The same two-part construction accommodates several choices of non-atom support by changing only the kernel $K(\cdot \mid \theta)$ and its base distribution. On the full real line one may use Normal kernels; on the positive real line, Gamma kernels parameterized by their component means; and on a bounded interval $[a, b]$, kernels defined after the affine transformation $x = (y-a)/(b-a) \in (0, 1)$. For a continuously observed bounded outcome this gives a density on the original scale through the usual Jacobian.

Bounded-score logit-Normal model. For a reported bounded score, let $\mathcal{S} = \{s_1, \ldots, s_J\} \subset [a, b]$ be the possible non-atom scores. The bounded-score model treats the recorded score as a discretized observation of an underlying bounded response rather than as an exact continuous measurement. A logit-Normal component with parameters $\theta_{rh} = (\mu_{rh}, \sigma_{rh})$ is defined by

$$X_{rh} \sim \text{LogitNormal}(\mu_{rh}, \sigma^2_{rh}), \qquad \text{logit}(X_{rh}) \sim N(\mu_{rh}, \sigma^2_{rh}), \qquad X_{rh} \in (0, 1),$$

with the corresponding response on the original scale given by $a + (b - a)X_{rh}$. This kernel is appropriate for bounded health scores because it assigns probability only to the admissible interval, can represent skewness and boundary-adjacent mass after transformation, and, through mixtures, can approximate irregular survivor-score distributions without imposing a single Gaussian shape on the original scale.

We define a heap as a candidate reporting-grid width q_ℓ, $\ell = 1, \ldots, L$. In the IST-3 analysis the heaps are $q_\ell \in \{1, 5, 10\}$ EQ-VAS points, corresponding to unit reporting and plausible five- or ten-point rounding. Score s_j is admissible on heap ℓ only if $s_j - a$ is an integer multiple of q_ℓ; otherwise its heap-specific probability is zero. If s_j is admissible on heap ℓ, observing s_j is interpreted as the event that the underlying bounded response falls in the centered rounding bin for that grid. Thus define

$$I_{\ell j} = [l_{\ell j}, u_{\ell j}] = \left[\frac{\max(a, s_j - q_\ell/2) - a}{b - a}, \frac{\min(b, s_j + q_\ell/2) - a}{b - a}\right],$$

and set $c_{\ell j} = 1$ for admissible score-heap pairs and $c_{\ell j} = 0$ otherwise. Let $G_\theta(x) = \Phi\{(\text{logit}\, x - \mu)/\sigma\}$ denote the logit-Normal CDF on the unit interval. The component probability of observing score s_j under heap ℓ is then

$$q_{rh\ell j}(\theta_{rh}) = c_{\ell j} \left\{G_{\theta_{rh}}(u_{\ell j}) - G_{\theta_{rh}}(l_{\ell j})\right\},$$

with endpoint intervals truncated to $[0, 1]$ and CDF values at 0 and 1 interpreted by continuity. If $\eta_{r\ell}$ are the heap weights in arm r, the arm-specific score probability is

$$p_{rj}(\vartheta) = \sum_{h=1}^{H} w_{rh} \sum_{\ell=1}^{L} \eta_{r\ell} q_{rh\ell j}(\theta_{rh}), \qquad \sum_{j=1}^{J} p_{rj}(\vartheta) = 1.$$

Thus, writing d_r for the number of atom observations in arm r, m_r for the number of non-atom observations, and n_{rj} for the number of reported non-atom scores equal to s_j, the concrete likelihood for the heaped reported-score model is, up to multinomial constants,

$$L_{\text{score}}(\vartheta) \propto \prod_{r=0}^{1} \rho_r^{d_r} \pi_r^{m_r} \prod_{j=1}^{J} p_{rj}(\vartheta)^{n_{rj}}, \qquad \pi_r = 1 - \rho_r.$$

Marginalizing over heaps accounts for reporting resolution directly in the likelihood, rather than treating all heaped scores as exact continuous observations. The heap weights may be constrained

to be common across arms, $\eta_{0\ell} = \eta_{1\ell}$, or allowed to vary by arm; either way the atom probability, mixture weights, and unit-interval kernel parameters retain the same interpretation. In the IST-3 application we use shared heap weights because the reporting grid is a feature of EQ-VAS measurement and recording, not a treatment-specific biological mechanism.

Writing $\widetilde{Y}_i^{(e_0)} = e_0$ when $A_i = 0$ and $\widetilde{Y}_i^{(e_0)} = Y_i$ when $A_i = 1$, the no-covariate model may be expressed hierarchically as

$$
\begin{aligned}
A_i \mid R_i = r &\sim \text{Bernoulli}(\pi_r), \qquad r \in \{0,1\},\\
\pi_r &= 1 - \rho_r, \qquad \rho_r \sim \text{Beta}(a_\rho, b_\rho),\\
Y_i \mid (A_i = 1, R_i = r) &\sim F_r^c,\\
F_r^c(dy) &= \sum_{h=1}^{H} w_{rh} K(dy \mid \theta_{rh}),\\
v_{rh} \mid \alpha_r &\sim \text{Beta}(1, \alpha_r), \qquad h = 1, \ldots, H-1,\\
\alpha_r &\sim \text{Gamma}(a_\alpha, b_\alpha),\\
w_{r1} &= v_{r1},\\
w_{rh} &= v_{rh} \prod_{\ell=1}^{h-1} (1 - v_{r\ell}), \qquad h = 2, \ldots, H-1,\\
w_{rH} &= \prod_{\ell=1}^{H-1} (1 - v_{r\ell}),\\
K(dy \mid \theta_{rh}) &= \phi(y; m_{rh}, s_{rh}^2)\, dy, \qquad \mathcal{C} = \mathbb{R},\\
m_{rh} &\sim N(m_0, s_0^2), \qquad s_{rh} \sim \text{Lognormal}(m_s, s_s^2),\\
K(dy \mid \theta_{rh}) &= g(y; a_{rh}, a_{rh}/m_{rh})\ dy, \qquad \mathcal{C} = (0, \infty),\\
m_{rh} &\sim \text{Lognormal}(m_m, s_m^2), \qquad a_{rh} \sim \text{Lognormal}(m_a, s_a^2),\\
K(\{s_j\} \mid \theta_{rh}, \eta_r) &= \sum_{\ell=1}^{L} \eta_{r\ell} c_{\ell j} \{G_{\theta_{rh}}(u_{\ell j}) - G_{\theta_{rh}}(l_{\ell j})\}, \qquad \mathcal{C} = \{s_1, \ldots, s_J\},\\
G_{\theta_{rh}}(x) &= I_x\{m_{rh}\phi_{rh}, (1 - m_{rh})\phi_{rh}\},\\
m_{rh} &\sim \text{Beta}(a_m, b_m), \qquad \phi_{rh} \sim \text{Lognormal}(m_\phi, s_\phi^2),\\
G_{\theta_{rh}}(x) &= \Phi\{(\text{logit}\, x - \mu_{rh})/\sigma_{rh}\}, \qquad \mu_{r1} < \cdots < \mu_{rH}, \qquad \sigma_{rh} > 0,\\
\log \sigma_{rh} &\sim N(m_\sigma, s_\sigma^2), \qquad \eta_r \sim \text{Dirichlet}(\zeta).
\end{aligned}
\tag{31}
$$

where $\phi(\cdot; m, s^2)$ denotes the Normal density with mean m and variance s^2, and $g(\cdot; a, b)$ denotes the Gamma density with shape a and rate b. Thus the full-real-line member uses Normal kernels, whereas the positive-support member uses Gamma kernels parameterized so that m_{rh} is the component mean. The latter formulation requires all non-atom outcomes to be strictly positive. For bounded outcomes, the unit-interval kernel may be either Beta or logit-Normal; for reported scores, the observed score probabilities are obtained by integrating the unit-interval kernel over the reporting intervals and averaging over the reporting grids. In all versions, the mixture is arm-specific, the atom probability is explicit, and the scientific summaries remain directly interpretable despite the flexibility of the non-atom component.

Posterior inference. Let $\mathcal{D}_n = \{(\widetilde{Y}_i^{(e_0)}, A_i, R_i) : i = 1, \ldots, n\}$ denote the observed data, and let ϑ collect all unknown quantities in Equation (31). For arm r, the posterior predictive distribution of a future numerical combined outcome $\widetilde{Y}_{r,\text{new}}^{(e_0)}$ is

$$\Pr\Big(\widetilde{Y}_{r,\text{new}}^{(e_0)} \in B \mid \mathcal{D}_n\Big) = \int \Bigg\{ \rho_r \delta_{e_0}(B) + \pi_r \int_{B \cap \mathcal{C}} f_r^c(y \mid \vartheta)\, \lambda_{\mathcal{C}}(dy) \Bigg\} p(\vartheta \mid \mathcal{D}_n)\, d\vartheta.$$

where

$$f_r^c(y \mid \vartheta) = \sum_{h=1}^{H} w_{rh} k(y \mid \theta_{rh})$$

is the arm-specific non-atom mixture density for continuous-support fits, with the analogous score probability mass function used for bounded-score fits. For posterior draw m, the corresponding mean among non-atom outcomes is

$$\mu_r^{c,(m)} = \int_{\mathcal{C}} y f_r^c(y \mid \vartheta^{(m)})\, \lambda_{\mathcal{C}}(dy),$$

with the integral replaced by $\sum_j s_j p_{rj}^{(m)}$ for bounded-score fits. Under the Normal and Gamma kernels this reduces to the weighted component-mean average $\mu_r^{c,(m)} = \sum_{h=1}^{H} w_{rh}^{(m)} m_{rh}^{(m)}$. The analysis reports the arm-specific summaries ρ_r, π_r, and μ_r^c together with the treatment contrasts

$$\begin{aligned} \Delta_{\text{atom}} &= \rho_1 - \rho_0, \qquad \mu_\delta = \mu_1^c - \mu_0^c, \qquad \alpha_\delta = \frac{\pi_1/(1-\pi_1)}{\pi_0/(1-\pi_0)}, \\ \mu_r^{(e_0)} &= e_0 \rho_r + \pi_r \mu_r^c, \qquad \Delta^{(e_0)} = \mu_1^{(e_0)} - \mu_0^{(e_0)}. \end{aligned} \tag{32}$$

For the atom-at-zero representation used elsewhere in the manuscript, $\Delta = \Delta^{(0)} = \pi_1 \mu_1^c - \pi_0 \mu_0^c$. Thus the Bayesian analysis reports, on a common posterior scale, the atom probability in each arm, the mean among non-atom outcomes in each arm, the atom-probability contrast, the contrast in non-atom means, the odds-ratio summary α_δ for escaping the atom, and the induced contrast on the numerical combined-outcome scale.

All of these functionals are computed draw-by-draw from the posterior sample and are then summarized by posterior medians or means, equal-tail credible intervals, and directional posterior probabilities. In the analyses below the default point summaries are posterior medians, with equal-tail credible intervals. Because several scientifically relevant quantities are nonlinear functions of (ρ_r, π_r, μ_r^c), it is important to compute them at each posterior draw rather than by plugging posterior means into the formulas in Equation (32). Directional summaries such as $\Pr(\Delta_{\text{atom}} < 0 \mid \mathcal{D}_n)$, $\Pr(\mu_\delta > 0 \mid \mathcal{D}_n)$, $\Pr(\alpha_\delta > 1 \mid \mathcal{D}_n)$, or $\Pr(\Delta^{(e_0)} > 0 \mid \mathcal{D}_n)$ are then immediate from the posterior draws, with the relevant direction chosen according to whether larger or smaller outcomes are clinically preferred.

Posterior predictive checks and model assessment. Model adequacy is assessed through posterior predictive checks in the sense of Gelman et al. (1996). These are Bayesian diagnostics based on the posterior predictive distribution, not frequentist goodness-of-fit tests. For each posterior draw $\vartheta^{(m)}$, one generates a replicated data set by first drawing

$$A_i^{\text{rep},(m)} \mid R_i = r, \vartheta^{(m)} \sim \text{Bernoulli}\Big(\pi_r^{(m)}\Big),$$

and then, for those replicated subjects with $A_i^{\mathrm{rep},(m)} = 1$,

$$Y_i^{\mathrm{rep},(m)} \mid \left(A_i^{\mathrm{rep},(m)} = 1, R_i = r, \vartheta^{(m)}\right) \sim F_r^{c,(m)}.$$

The replicated combined outcome is finally reconstructed as $\widetilde{Y}_i^{\mathrm{rep},(e_0,m)} = e_0$ when $A_i^{\mathrm{rep},(m)} = 0$ and $\widetilde{Y}_i^{\mathrm{rep},(e_0,m)} = Y_i^{\mathrm{rep},(m)}$ otherwise. The observed data and the replicated data are then compared through discrepancies chosen to target the two components of the model.

For the atom part, the relevant question is whether the fitted Bernoulli model reproduces the observed armwise frequency of the atom. A natural discrepancy is therefore the maximum armwise absolute deviation between the observed atom rate and the fitted atom probability,

$$T_A(\widetilde{y}, \vartheta) = \max_{r \in \{0,1\}} \left| \frac{1}{n_r} \sum_{i: R_i = r} \mathbb{1}\{\widetilde{y}_i = e_0\} - \rho_r \right|.$$

For the continuous non-atom part, the relevant question is whether the fitted mixture reproduces the shape of the conditional distribution among subjects with observed outcomes. The implementation therefore uses the maximum armwise Kolmogorov–Smirnov discrepancy between the empirical non-atom distribution and the fitted continuous mixture CDF,

$$T_C(y, \vartheta) = \max_{r \in \{0,1\}} \sup_{t \in \mathcal{C}} \left| \widehat{F}_{r,\mathrm{obs}}^c(t) - F_r^c(t \mid \vartheta) \right|.$$

For bounded reported-score models, the same diagnostic is evaluated as a maximum armwise discrepancy between the empirical score CDF and the fitted discrete score CDF. For positive-support models the same comparison can be displayed on the log scale for graphical stability, but the target remains the arm-specific conditional non-atom law. These checks are supplemented naturally by graphical posterior predictive displays, such as grouped bar plots for the atom indicators and density, mass, or CDF overlays for the non-atom outcomes.

More generally, one may also examine scalar posterior predictive discrepancies based on arm-specific means, tail probabilities, quantiles, or other summaries chosen for the application. For any discrepancy $T(\cdot, \vartheta)$, the posterior predictive p-value is

$$p_{\mathrm{ppc}} = \Pr\left\{ T(\widetilde{Y}^{\mathrm{rep}}, \vartheta) \geq T(\widetilde{Y}, \vartheta) \,\middle|\, \mathcal{D}_n \right\},$$

estimated by the corresponding empirical proportion over posterior draws and replicated data sets. Such quantities are useful diagnostics for systematic lack of fit in either the atom model or the continuous conditional distribution, but they should be interpreted as Bayesian model-checking summaries rather than as classical significance tests.

Covariate extensions. The no-covariate treatment-stratified model is the primary development in this paper, but the same two-part architecture extends naturally when baseline covariates are present. A direct extension retains treatment stratification and replaces the arm-specific constants π_r and F_r^c by covariate-dependent quantities $\pi_r(x)$ and $F_r^c(\cdot \mid x)$. For example, one may write

$$A_i \mid R_i = r, X_i = x \sim \mathrm{Bernoulli}\big(\pi_r(x)\big),$$
$$Y_i \mid (A_i = 1, R_i = r, X_i = x) \sim F_r^c(\cdot \mid x),$$

and then let the component-specific kernel means vary with x within each treatment arm. A convenient dependent Dirichlet process construction is

$$Y_i \mid (A_i = 1, R_i = r, X_i = x, z_i = h) \sim K(\cdot \mid m_{rh}(x), \vartheta_{rh}),$$
$$m_{rh}(\cdot) \sim \mathcal{GP}(\mu_{0h}(\cdot), C_h(\cdot, \cdot)),$$

so that the kernel locations vary smoothly with the covariates while the random mixing distribution remains treatment-arm specific. This yields a predictor-dependent mixture model in the spirit of dependent Dirichlet process constructions (MacEachern, 1999; Gelfand et al., 2005); see also Rasmussen and Williams (2006) for Gaussian process priors. Such a formulation preserves the two-part interpretation while allowing treatment groups to differ not only in atom probability but also in how the entire non-atom distribution evolves over the covariates.

If a more modular regression component is preferred, the Gaussian-process mean functions can be replaced by Bayesian additive regression trees on the kernel means, thereby retaining a flexible regression surface while leaving the mixture structure unchanged (Chipman et al., 2010). This is attractive when nonlinear covariate effects and interactions are important but a fully predictor-dependent random measure is unnecessary.

When a covariate is purely categorical, especially when it indexes substantively related groups, a hierarchical construction is often more appropriate than a fully continuous predictor-dependent process. For levels $\ell = 1, \ldots, L$ of a factor covariate, one may introduce within each treatment arm

$$G_r^\star \sim \mathrm{DP}(\gamma_r, H_0), \qquad G_{r\ell} \mid G_r^\star \sim \mathrm{DP}(\alpha_r, G_r^\star),$$

so that

$$F_{r\ell}^c(B) = \int_\Theta K(B \mid \theta)\, G_{r\ell}(d\theta), \qquad B \subseteq \mathcal{C}.$$

This is a hierarchical Dirichlet process mixture model (Teh et al., 2006); ANOVA-type dependent random-measure formulations provide a closely related alternative for factor effects (De Iorio et al., 2004). If the factor levels correspond to different hospital sites, such a hierarchical construction is especially natural. Hospitals are expected to have related but not identical outcome distributions because of differences in case mix, clinical practice, or measurement processes. The site-specific random measures $G_{r\ell}$ therefore allow each hospital to retain its own atom probability model and its own non-atom distribution, while the shared parent measure $G_r^\star$ induces borrowing of strength by encouraging the hospitals within a treatment arm to reuse common mixture atoms. In this way, data-rich sites stabilize inference for data-sparse sites without forcing the site-specific distributions to be identical. These covariate-adjusted extensions are conceptually straightforward continuations of the no-covariate framework, but the basic two-group model already captures the main ideas needed for the present paper.

3 Simulation study

The simulation study had two purposes: to assess null calibration and to compare power with standard one-dimensional analyses when treatment acts through the atom, the positive part, or both. In every replicate, the two treatment groups had equal size, we worked on the atom-at-zero numerical representation, equivalently the coded variable $\widetilde{Y}_i^{\,0} = c_0(\widetilde{Y}_i)$, and each subject contributed the numerical combined outcome $\widetilde{Y}_i^{\,0} = A_i Y_i$, where $A_i \mid R_i = r$ was Bernoulli with scenario-specific probability π_r and $Y_i \mid (A_i = 1, R_i = r)$ followed the corresponding survivor distribution.

Table 1 lists the data-generating mechanisms. The scenarios were chosen to isolate the practically important comparisons: a low-atom normal reference case (Scenario 1), a pure atom-probability effect (Scenario 2), an antagonistic case in which the atom and continuous components move in opposite directions while the mean of the coded combined outcome stays fixed (Scenario 3), a concordant joint improvement case (Scenario 4), a gamma shape-change case with survivor-mean shift (Scenario 5), and a rare-outlier contamination case that is deliberately favorable to rank-based methods (Scenario 6).

Table 1: Simulation scenarios for the TruncComp2 power study

Scenario	Configuration	$P(A=1 \mid R)$	$Y \mid A=1, R$
S1	Normal survivor shift, low atom	$\pi_0=\pi_1=0.85$	$Y \mid A=1, R=0 \sim N(3,1)$, $Y \mid A=1, R=1 \sim N(3+\delta_h,1)$, $\delta_h \in \{0, 0.20, 0.35, 0.50\}$
S2	Survival-only effect	$\pi_0=0.45$, $\pi_1 \in \{0.45, 0.50, 0.55, 0.60\}$	$Y \mid A=1, R=0 \sim Y \mid A=1, R=1 \sim N(3,1)$
S3	Antagonistic cancellation	$\pi_0=0.50$, $\pi_1 \in \{0.50, 0.525, 0.55, 0.60\}$	$Y \mid A=1, R=0 \sim N(3,1)$, $Y \mid A=1, R=1 \sim N(1.5/\pi_1,1)$
S4	Concordant joint improvement	$\pi_0=0.50$, $\pi_1 \in \{0.50, 0.55, 0.60, 0.65\}$	$Y \mid A=1, R=0 \sim N(3,1)$, $Y \mid A=1, R=1 \sim N(3+0.15h,1)$
S5	Gamma shape-change with mean shift	$\pi_0=\pi_1=0.65$	$Y \mid A=1, R=0 \sim \Gamma(9,3/9)$, $Y \mid A=1, R=1 \sim \Gamma(k_h, m_h/k_h)$, $(k_h, m_h) \in \{(9,3.0),(6,3.2),(4,3.4),(2.5,3.6)\}$
S6	Rare-outlier contamination	$\pi_0=\pi_1=0.80$	$Y \mid A=1, R=r$ is $0.95N(3+\delta_h r, 0.35^2)+0.05N(10+\delta_h r, 0.5^2)$, $\delta_h \in \{0, 0.15, 0.25, 0.35\}$

For each scenario we considered effect levels $h \in \{0,1,2,3\}$, with $h=0$ as the null configuration, and per-group sample sizes $n \in \{50, 100, 150, 200, 300, 400, 500\}$. This yielded $6 \times 4 \times 7 = 168$ scenario-effect-sample-size cells, and each cell used 25,000 Monte Carlo repetitions. Power and empirical size were estimated as rejection proportions at the 5% level, so the worst-case Monte Carlo standard error of an estimated rejection rate was approximately 0.0032. We compared the parametric and semi-parametric likelihood-ratio tests with Welch's two-sample t-test and the Wilcoxon rank-sum test applied directly to the combined outcome; Appendix A records the exact simulation protocol and implementation details needed to reproduce the study.

Figure 1 shows power as a function of sample size at the strongest non-null level across all six scenarios. Figure 2 complements that display by plotting power as a function of the effect level at $n=150$ observations per group across the same six scenarios. The supplementary material reports the full scenario-specific power tables for all 168 cells and the null-calibration displays for the $h=0$ settings.

The results support the main motivation for the proposed tests, but they also show that no single procedure is uniformly best. Scenario 1 is the reference case for the parametric assumptions: the survivor distribution is normal, the atom probability is unchanged, and treatment shifts the survivor mean. At $h=3$ and $n=150$, the likelihood-ratio tests rejected 95.4% of replicates, compared with 85.4% for Wilcoxon and 69.4% for the t-test. In Scenario 2, where treatment acts only through the atom probability, all four procedures detected the effect similarly. The one-dimensional tests were slightly ahead at $n=150$ (66.1% for Wilcoxon versus 64.4% for either likelihood-ratio test), and by $n=500$ all four rejection rates were about 99%.

Scenario 3 is the clearest verification of the two-component idea. In that antagonistic setting, the coded combined-outcome mean is fixed at every effect level even though the atom probability and survivor mean both change. At $h=3$ and $n=150$, the likelihood-ratio tests reached about 91.5% power, whereas the Wilcoxon and t-tests remained near their null rejection rates. By $n=500$, both likelihood-ratio procedures rejected every replicate, while the one-dimensional tests still had little signal. Scenario 4 is the least discriminating non-null scenario for comparing methods: when both components improve concordantly, the effect is already visible on the coded combined-outcome scale, and all four methods have similar power.

The non-normal scenarios illustrate the role of the survivor distribution. In Scenario 5, the survivor distribution changes shape and mean. The semi-parametric likelihood-ratio test is consistently

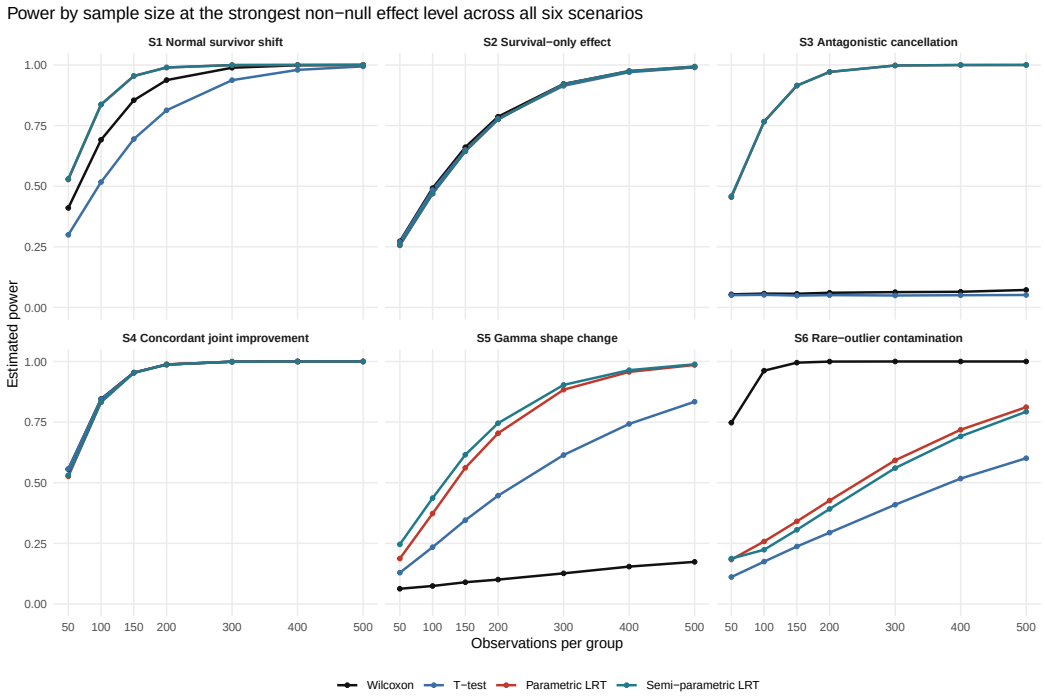

Figure 1: Estimated power as a function of sample size at the strongest non-null level for Scenarios 1 through 6, displayed in a 2×3 layout.

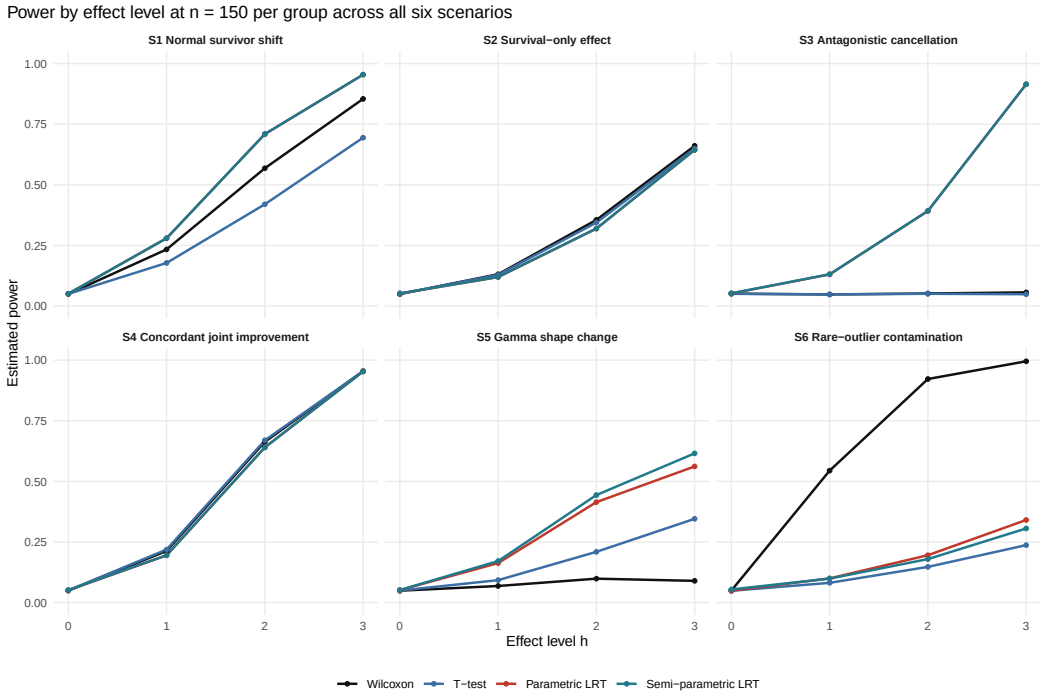

Figure 2: Estimated power as a function of effect level at $n = 150$ observations per group for Scenarios 1 through 6, displayed in a 2×3 layout.

the most powerful procedure, the parametric likelihood-ratio test is close but lower, the t-test is intermediate, and Wilcoxon has poor power; at $h = 3$ and $n = 150$, the corresponding rejection rates were 61.5%, 56.2%, 34.6%, and 9.0%. Scenario 6 deliberately favors a rank test through rare outlier contamination. Wilcoxon therefore dominates, while the likelihood-ratio tests outperform the t-test but are not competitive with Wilcoxon; at $h = 3$ and $n = 150$, Wilcoxon rejected 99.5% of replicates, compared with 34.1% for the parametric likelihood-ratio test, 30.6% for the semi-parametric likelihood-ratio test, and 23.7% for the t-test.

The null-calibration results are mostly reassuring but not perfectly uniform. Across the full simulation run, all method failure rates were zero. Most null rejection rates were close to the nominal 5% level for all four methods. The important exception was the semi-parametric likelihood-ratio test in the rare-outlier null setting: its rejection rate was 12.6% at $n = 50$ and 6.6% at $n = 100$, before settling near nominal from $n \geq 150$. Thus the main conclusions are strongest for moderate sample sizes, while very small contaminated samples require caution for the semi-parametric procedure. The long supplementary tables document the full scenario-by-sample-size patterns underlying Figure 1.

4 Clinical application: IST-3 stroke trial

We apply the proposed methods to the third International Stroke Trial (IST-3), a randomized trial of intravenous alteplase (rt-PA) versus control in acute ischemic stroke (IST-3 Collaborative Group, 2012; Sandercock et al., 2016). Acute ischemic stroke is a natural setting for atom-plus-continuous endpoints: treatment may affect both mortality and functional health, and death before follow-up is not a missing quality-of-life value but a clinically meaningful terminal state. Among participants alive at 6 months with observed EQ-VAS, the EuroQol visual analogue score is a patient-centered measure on a 0 to 100 scale, with larger values indicating better self-rated health.

The IST-3 data were downloaded from Edinburgh DataShare. The fixed-width data file `ist3.dat` and accompanying SAS syntax identify allocated treatment, death by 6 months, and 6-month EQ-VAS. The syntax labels `itt_treat` as allocated treatment, `dead6mo` as death before 6 months, and `euroqol6` as the 6-month EuroQol health state. We verified the treatment coding against the text treatment field: `itt_treat = 0` corresponds to rt-PA and `itt_treat = 1` corresponds to placebo/control. We therefore set $R = 1$ for rt-PA and $R = 0$ for placebo/control. The DataShare record is publicly downloadable after the end of the embargo, but reuse is conditional on the IST-3 Data Use Agreement; the data citation is Sandercock et al. (2016), DOI 10.7488/ds/1350.

For the application endpoint, we code

$$Y = \begin{cases} -1, & \text{death by 6 months,} \\ \text{6-month EQ-VAS,} & \text{alive with observed EQ-VAS,} \end{cases} \qquad A = \mathbb{1}(Y \neq -1).$$

Thus the atom is -1, representing death by 6 months, and the non-atom component is EQ-VAS for participants alive with an observed 6-month score. Alive participants with missing 6-month EQ-VAS were excluded from this analysis because the non-atom component was not observed; deaths were retained through the atom. This left 2,733 participants for analysis, including 1,361 placebo/control participants and 1,372 rt-PA participants. Table 2 reports the analysis sample, death counts, EQ-VAS summaries among those alive with observed scores, and the combined endpoint mean.

Table 2: IST-3 analysis sample and endpoint summaries.

Summary	Placebo/control	rt-PA
Randomized participants	1,520	1,515
Included in analysis	1,361	1,372
Alive but missing 6-month EQ-VAS	159	143
Death atom, n (%)	407 (29.9%)	408 (29.7%)
Alive with EQ-VAS, n (%)	954 (70.1%)	964 (70.3%)
EQ-VAS, alive with observed score, mean (SD)	57.9 (23.8)	60.7 (23.3)
EQ-VAS, alive with observed score, median (IQR)	60 (40, 75)	63 (49, 80)
EQ-VAS, alive with observed score, range	0 to 100	0 to 100
Combined endpoint mean	40.3	42.3

Figure 3 displays the atom and non-atom components together with the semiparametric joint confidence region. The death atom was similar in the two arms: 407 placebo/control participants (29.9%) and 408 rt-PA participants (29.7%) died before 6 months. Among participants alive with observed EQ-VAS, however, the mean was 57.9 in the placebo/control arm and 60.7 in the rt-PA arm, giving $\hat{\mu}_\delta = 2.82$ points. The fitted odds ratio for being alive with observed EQ-VAS rather than at the death atom was $\hat{\alpha}_\delta = 1.008$, and the combined mean contrast under the $Y = -1$ death coding was $\hat{\Delta} = 2.08$.

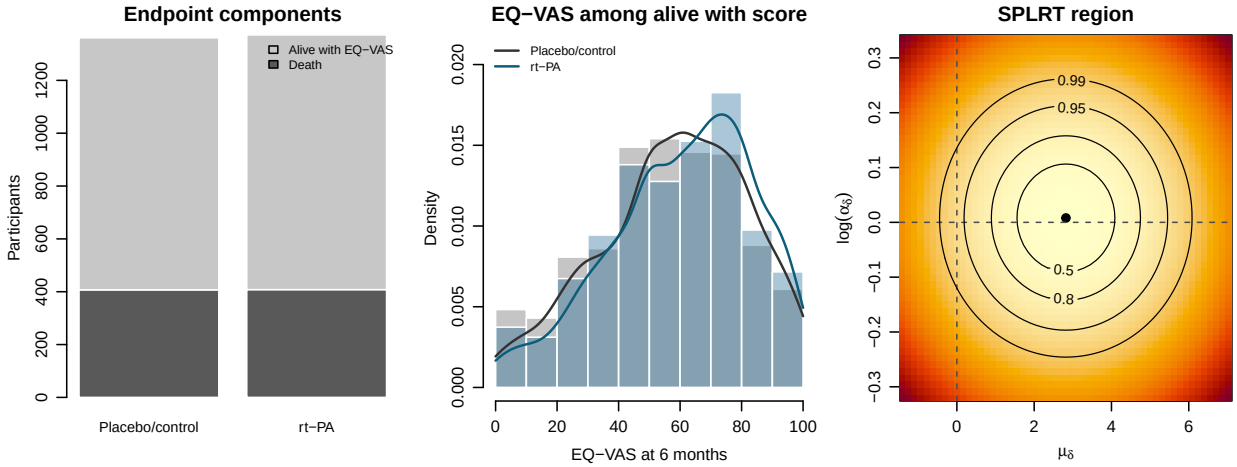

Figure 3: IST-3 endpoint components and semiparametric likelihood-ratio region. The left panel shows death and alive-with-EQ-VAS counts by treatment arm. The middle panel shows the EQ-VAS distribution among participants alive with observed scores. The right panel shows the semiparametric likelihood-ratio surface for $(\mu_\delta, \log \alpha_\delta)$ with reference lines at no non-atom mean difference and no atom-probability difference.

Table 3 compares standard one-dimensional analyses with the proposed joint tests. A Welch t-test applied directly to the atom-coded endpoint gave p-value 0.109, and the Wilcoxon rank-sum test gave p-value 0.094. The atom-coded t-test depends on the numerical choice -1 for death; moving the death atom would change the mean contrast and its standard error. The Wilcoxon test is less tied to that exact value, but it treats all deaths as tied observations at the worst rank and still collapses the endpoint to one dimension. An analysis restricted to participants alive with observed EQ-VAS is more sensitive to the EQ-VAS difference, but it conditions on a post-randomization

alive-at-follow-up event and therefore answers a different observed-data question.

Table 3: Standard and likelihood-ratio analyses of the IST-3 endpoint.

Analysis	Method	Estimate/target	Statistic	p-value
Combined endpoint	Welch t-test	$\Delta = 2.08$	-1.60	0.109
Combined endpoint	Wilcoxon rank-sum	Death tied as worst value	899,629.50	0.094
Alive with EQ-VAS	Welch t-test	$\mu_\delta = 2.82$	-2.62	0.009
Alive with EQ-VAS	Wilcoxon rank-sum	Conditional on being alive with observed EQ-VAS	427,369.50	0.007
Death atom	Fisher exact test	Death proportions		0.933
Parametric model	Joint likelihood-ratio test	$\mu_\delta = 2.82$, $\alpha_\delta = 1.008$, $\Delta = 2.08$	6.88	0.032
Semiparametric model	Joint likelihood-ratio test	$\mu_\delta = 2.82$, $\alpha_\delta = 1.008$, $\Delta = 2.08$	6.87	0.032

The likelihood-ratio procedures test the joint null of no treatment effect on the death atom probability and no treatment effect on non-atom EQ-VAS. The parametric LRT statistic was 6.88 with p-value 0.032, and the semiparametric LRT statistic was 6.87 with p-value 0.032. In this data set the two procedures agree closely, suggesting that the signal is mainly a location shift in the EQ-VAS component rather than a feature driven by the parametric distributional assumption. Clinically, μ_δ is the rt-PA minus control mean EQ-VAS difference among participants alive with observed scores; α_δ is the rt-PA/control odds ratio for being alive with observed EQ-VAS rather than at the death atom; and Δ is the rt-PA minus control mean of the combined endpoint after assigning death the value -1. The joint rejection should therefore be interpreted as evidence that the observed two-component endpoint distribution differs between randomized arms, not as a principal-stratum causal effect.

Table 4: Confidence intervals for the combined mean contrast Δ from the semiparametric IST-3 fit.

Interval construction	Lower	Upper	Status
Welch	-0.46	4.63	Estimated
Profile likelihood	-0.46	4.62	Estimated
Projected joint region	-1.09	5.25	Estimated

Table 4 illustrates three confidence intervals for Δ implemented in the software. The Welch interval is the direct interval for the atom-coded mean contrast. The profile interval constrains the full likelihood at a fixed Δ, while the projected interval maps the joint confidence region for the component parameters into the implied combined mean contrast. In IST-3, the profile interval is nearly identical to the Welch interval, whereas the projected interval is wider because it preserves joint uncertainty in both the atom and non-atom components.

We also fit the package Bayesian two-part model using the bounded-score logit-Normal likelihood defined in Section 2.4. The Bernoulli atom probabilities used independent $\text{Beta}(1, 1)$ priors. The non-atom EQ-VAS distribution used $H = 2$ mixture components, score range 0 to 100, unit score spacing, ordered component locations on the logit scale, and shared heap weights over 1-, 5-, and 10-point reporting grids. The stick-breaking concentration parameters used $\text{Gamma}(4, 4)$ priors, the logit-scale component locations used $\text{Normal}(0, 2)$ priors, the component scales used $\text{Lognormal}(\log 1.2, 0.5)$ priors, and the heap weights used a $\text{Dirichlet}(2, 2, 2)$ prior. The manuscript run used four chains with 1,000 warmup and 2,000 post-warmup iterations per chain at `adapt_delta = 0.995`. The fit is cached with the generated application data, and

setting `TRUNCCOMP_REFRESH_IST3_BAYES=true` regenerates it from the fixed-width file. Because this Bayesian implementation uses a practical finite mixture for the manuscript build, we report sampler and mixture-weight diagnostics alongside the posterior summaries.

Table 5: Bayesian IST-3 posterior summaries from the bounded-score logit-normal two-part mixture model.

Quantity	Posterior median	95% CrI	Probability
Death probability, control	0.299	0.28 to 0.32	
Death probability, rt-PA	0.297	0.27 to 0.32	
Alive/EQ-VAS probability, control	0.701	0.68 to 0.72	
Alive/EQ-VAS probability, rt-PA	0.703	0.68 to 0.73	
Mean EQ-VAS, control	57.48	56.04 to 58.92	
Mean EQ-VAS, rt-PA	60.22	58.74 to 61.66	
Death risk difference	-0.002	-0.04 to 0.03	$\Pr(< 0) = 0.541$
μ_δ	2.74	0.63 to 4.79	$\Pr(> 0) = 0.995$
α_δ	1.009	0.85 to 1.19	$\Pr(> 1) = 0.541$
Δ	2.03	-0.48 to 4.58	$\Pr(> 0) = 0.942$
Sampler diagnostics	–	–	Divergences 0; max $\hat{R}$ 1.001; min bulk ESS 7,915
Mixture-weight diagnostics	–	–	max $\hat{R}$ 1.000; min bulk ESS 4,136; passed = yes

The posterior summaries in Table 5 separate the atom/alive component from the non-atom EQ-VAS component. The benefit-oriented combined endpoint uses the same coding as the frequentist analysis, so $\Pr(\Delta > 0 \mid \text{data})$ is the posterior probability that rt-PA improves the combined endpoint. The posterior probabilities $\Pr(\mu_\delta > 0 \mid \text{data})$ and $\Pr(\alpha_\delta > 1 \mid \text{data})$ separately summarize benefit on EQ-VAS among participants alive with observed scores and on avoiding the atom. Figure 4 shows the posterior densities for these component and combined contrasts.

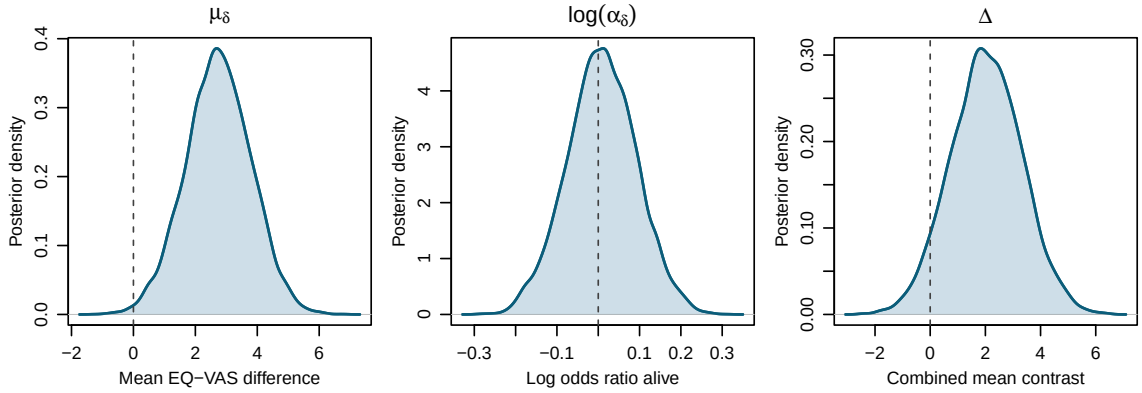

Figure 4: Posterior distributions for the IST-3 treatment contrasts. Vertical dashed lines mark no non-atom mean difference, no difference in the odds of being alive with observed EQ-VAS, and no combined endpoint mean difference.

Finally, we assessed whether the Bayesian model reproduced both parts of the endpoint. Table 6 reports posterior predictive checks for the atom model and the non-atom EQ-VAS distribution, and Figure 5 displays the corresponding replicated data summaries. These checks are important because a model can match the death proportions while still misrepresenting the bounded, heaped, and potentially skewed EQ-VAS distribution among participants alive with observed scores. The

reported-score check therefore compares replicated and observed non-atom distributions as a model check, not as evidence for treatment efficacy.

Table 6: Posterior predictive checks for the IST-3 Bayesian model.

Component	Statistic	Posterior predictive p	Notes
Death atom	Maximum armwise absolute atom-rate deviation	0.545	200 draws; Indicator for Y = atom scale
EQ-VAS reported scores	Maximum armwise discrete CDF discrepancy	0.325	200 draws; Reported score scale

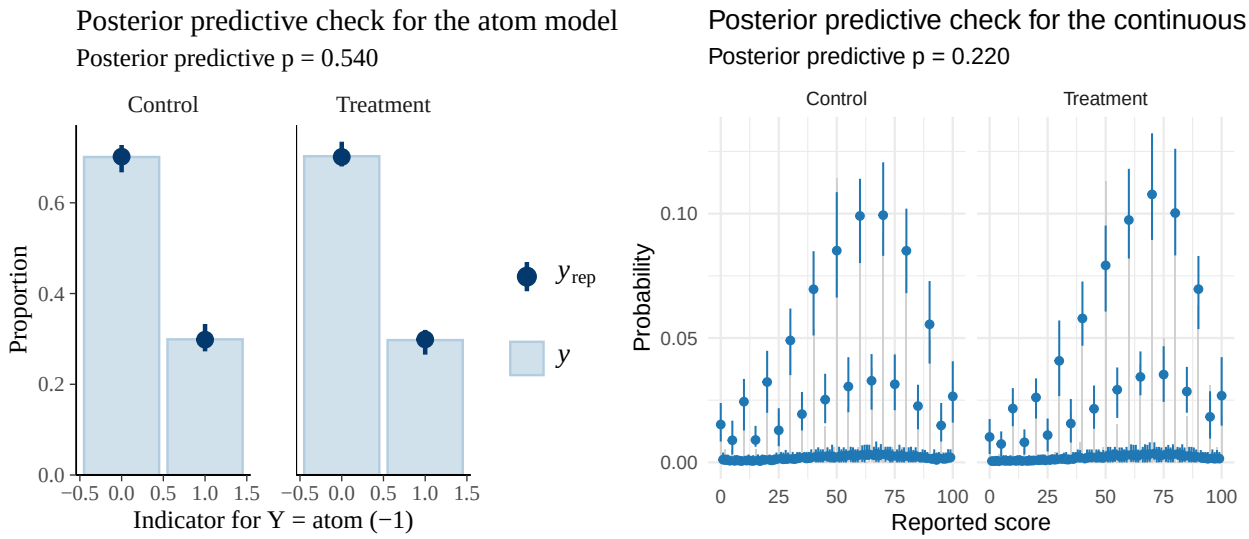

Figure 5: Posterior predictive checks for the IST-3 Bayesian model, separately for the death atom and the non-atom EQ-VAS component.

5 Discussion

We have proposed parametric and semi-parametric likelihood-ratio tests for continuous outcomes with a clinically meaningful atom. The key step is to retain the atomized outcome used in practice while testing the joint null of no treatment effect on the atom probability and no treatment effect on the positive part. This yields one prespecified p-value together with interpretable component-wise effect estimates and confidence regions.

The simulations show why that distinction matters. The clearest gain appears when the atom and positive part move in opposite directions, because that is the setting in which a one-dimensional composite can cancel real component-level signal. When only the atom probability changes or when both components move concordantly, one-dimensional tests can be similarly powerful. The IST-3 application serves the same purpose in a clinically recognizable setting: the joint tests detect two-component structure that is only weakly visible to standard analyses of the raw combined outcome.

This observed-data focus should also guide interpretation. The parameter pair $(\mu_\delta, \beta_\delta)$ is not a

survivor-average causal effect or a principal-stratum estimand; it describes the two components of the observed outcome distribution under randomization. The procedures are asymptotic, require positive numbers of observed outcomes in both groups, and the parametric version is only as credible as the model specified for $Y \mid A = 1$. The semi-parametric version relaxes that distributional assumption, but it still relies on regularity of the Bernoulli model and on feasible empirical-likelihood constraints.

For trials that require a single prespecified primary analysis, the framework offers one joint p-value together with clinically interpretable follow-up summaries for the binary and continuous components. That makes it attractive when investigators want to preserve the meaning of a worst-rank or atom-at-zero endpoint without collapsing the scientific question to a single distribution on the combined scale.

Software implementing both procedures, including the code used to generate the manuscript analyses, is available on GitHub (Jensen and Lange, 2018).

A Supplementary simulation results

This appendix collects the simulation figures and tables referenced in Section 3. It also records the exact simulation protocol implemented in `simulation-study/scripts/run-simulation-study.R` and `simulation-study/R/simulation-study.R`.

The simulation study used the full factorial design generated by the code defaults: six scenarios, four effect levels $h \in \{0, 1, 2, 3\}$, and seven per-group sample sizes $n \in \{50, 100, 150, 200, 300, 400, 500\}$, for a total of 168 cells. Each replicate contained exactly n control observations and n treatment observations. Within group $r \in \{0, 1\}$, the indicator A of whether the continuous outcome was observed was drawn as $A \sim \text{Bernoulli}(\pi_r)$. Conditional on $A = 1$, the positive outcome was drawn from a scenario-specific distribution $F_{r,h}$; conditional on $A = 0$, the numerical atom value was recorded as 0. Thus the analyzed endpoint was always the atom-at-zero numerical combined outcome $\widetilde{Y}^0 = AY$, equivalently $c_0(\widetilde{Y})$, and no covariate adjustment was used.

The six scenario families were as follows. Scenario 1 used $\pi_0 = \pi_1 = 0.85$, $Y \mid A = 1, R = 0 \sim N(3, 1)$, and $Y \mid A = 1, R = 1 \sim N(3 + \delta_h, 1)$ with $\delta_h \in \{0, 0.20, 0.35, 0.50\}$. Scenario 2 used $\pi_0 = 0.45$, $\pi_1 \in \{0.45, 0.50, 0.55, 0.60\}$, and $Y \mid A = 1, R = 0 \sim Y \mid A = 1, R = 1 \sim N(3, 1)$. Scenario 3 used $\pi_0 = 0.50$, $\pi_1 \in \{0.50, 0.525, 0.55, 0.60\}$, $Y \mid A = 1, R = 0 \sim N(3, 1)$, and $Y \mid A = 1, R = 1 \sim N(1.5/\pi_1, 1)$, so that $E(\widetilde{Y}^0 \mid R = 1) = E(\widetilde{Y}^0 \mid R = 0) = 1.5$ at every effect level even though both components change. Scenario 4 used $\pi_0 = 0.50$, $\pi_1 \in \{0.50, 0.55, 0.60, 0.65\}$, $Y \mid A = 1, R = 0 \sim N(3, 1)$, and $Y \mid A = 1, R = 1 \sim N(3 + 0.15h, 1)$. Scenario 5 used $\pi_0 = \pi_1 = 0.65$, $Y \mid A = 1, R = 0 \sim \Gamma(9, 3/9)$, and $Y \mid A = 1, R = 1 \sim \Gamma(k_h, m_h/k_h)$ with $(k_h, m_h) \in \{(9, 3.0), (6, 3.2), (4, 3.4), (2.5, 3.6)\}$, where $\Gamma(k, \theta)$ denotes a gamma distribution with shape k and scale θ. Scenario 6 used $\pi_0 = \pi_1 = 0.80$ and $Y \mid A = 1, R = r \sim 0.95\, N(3+\delta_h r, 0.35^2) + 0.05\, N(10 + \delta_h r, 0.5^2)$ with $\delta_h \in \{0, 0.15, 0.25, 0.35\}$.

For each cell, the driver called `set.seed(20260416 + cell_id - 1)` before generating the 25,000 replicates, where `cell_id` increases first over scenarios, then over effect levels, and then over sample sizes. Each replicate was analyzed by four methods applied to the combined outcome data frame produced by `simulateTruncatedData()`: `wilcox.test(Y ~ R, exact = FALSE)` for the Wilcoxon rank-sum test, `t.test(Y ~ R)` for Welch's two-sample t-test, `truncComp(Y ~ R, atom = 0, method = "LRT")` for the parametric likelihood-ratio test, and `truncComp(Y ~ R, atom = 0, method = "SPLRT")` for the semi-parametric likelihood-ratio test. In the package implementation, the parametric test combines a logistic likelihood-ratio comparison for A with a Gaussian likelihood-ratio comparison for Y among observations with $A = 1$, whereas the semi-parametric test combines the same logistic component with an empirical-likelihood test for the survivor mean difference. Both proposed procedures report a two-degree-of-freedom χ^2-calibrated joint p-value.

Within each cell, rejection rates were computed as the proportion of successful replicates with p-value below 0.05 for each method, and the Monte Carlo standard error was computed from that rejection proportion using the number of successful replicates for the given method. If a method call errored or returned a non-finite p-value, that replicate was counted as a failure for that method. The aggregated results object stores method-specific failure rates and runtimes, and the supplementary tables report the largest failure rate across the four methods in each cell. For the manuscript displays, Figure 1 uses the strongest non-null effect level ($h = 3$) across all six scenarios; Figure 2 uses all four effect levels at $n = 150$ per group across all six scenarios; and Figure 6 uses the null cells with $h = 0$ across all six scenarios. The six long supplementary tables list the complete power results for all 168 cells together with the expected coded combined-outcome

contrast $E(\widetilde{Y}^0 \mid R = 1) - E(\widetilde{Y}^0 \mid R = 0)$ used in the code-generated summaries.

Figure 6 summarizes the $h = 0$ null-calibration results across all six scenarios.

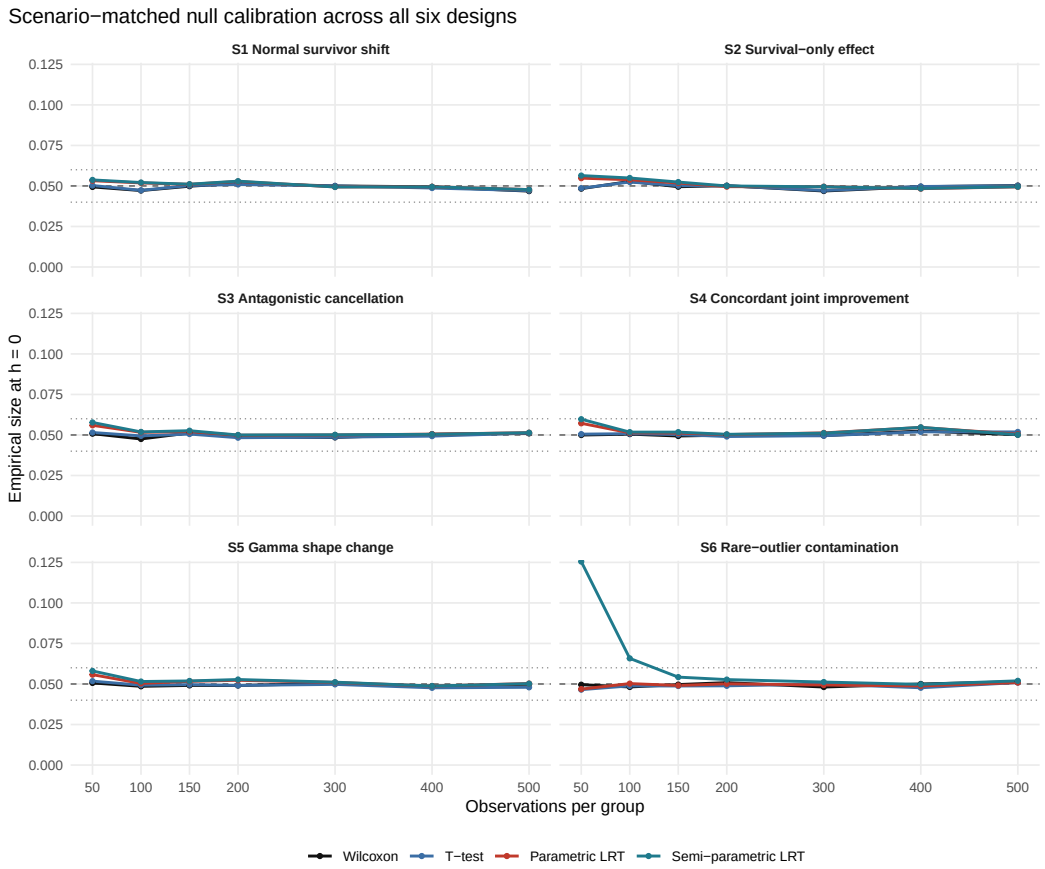

Figure 6: Empirical size at effect level $h = 0$ across all six simulation scenarios. The dashed line marks 0.05 and the dotted lines mark 0.04 and 0.06.

Table 7: Normal survivor shift, low atom: estimated power by sample size and effect level

n	h	Effect	Exp. Δ	Wilcoxon	T-test	LRT	SPLRT	Max fail
50	0	$\delta = 0.00$	0.000	4.94	5.03	5.32	5.37	0
100	0	$\delta = 0.00$	0.000	4.72	4.73	5.19	5.21	0
150	0	$\delta = 0.00$	0.000	4.98	5.02	5.10	5.11	0
200	0	$\delta = 0.00$	0.000	5.16	5.09	5.28	5.30	0
300	0	$\delta = 0.00$	0.000	5.00	5.00	4.94	4.94	0
400	0	$\delta = 0.00$	0.000	4.95	4.87	4.95	4.94	0
500	0	$\delta = 0.00$	0.000	4.68	4.71	4.78	4.78	0
50	1	$\delta = 0.20$	0.170	10.65	9.25	12.44	12.56	0
100	1	$\delta = 0.20$	0.170	16.89	13.26	20.16	20.17	0
150	1	$\delta = 0.20$	0.170	23.37	17.78	27.97	27.97	0
200	1	$\delta = 0.20$	0.170	29.32	21.86	35.98	36.01	0
300	1	$\delta = 0.20$	0.170	40.84	30.20	50.76	50.78	0
400	1	$\delta = 0.20$	0.170	52.36	39.04	64.54	64.60	0
500	1	$\delta = 0.20$	0.170	60.72	45.90	74.45	74.40	0
50	2	$\delta = 0.35$	0.298	22.98	17.55	28.78	29.02	0
100	2	$\delta = 0.35$	0.298	41.03	30.20	51.88	52.00	0
150	2	$\delta = 0.35$	0.298	56.84	41.98	70.91	70.96	0
200	2	$\delta = 0.35$	0.298	69.77	53.36	83.26	83.22	0
300	2	$\delta = 0.35$	0.298	85.27	69.76	95.02	95.03	0

Table 7: Normal survivor shift, low atom: estimated power by sample size and effect level *(continued)*

n	h	Effect	Exp. Δ	Wilcoxon	T-test	LRT	SPLRT	Max fail
400	2	$\delta = 0.35$	0.298	93.56	81.58	98.80	98.80	0
500	2	$\delta = 0.35$	0.298	97.37	89.66	99.76	99.76	0
50	3	$\delta = 0.50$	0.425	41.04	29.95	52.75	53.05	0
100	3	$\delta = 0.50$	0.425	69.14	51.71	83.67	83.70	0
150	3	$\delta = 0.50$	0.425	85.45	69.44	95.45	95.44	0
200	3	$\delta = 0.50$	0.425	93.76	81.30	98.95	98.95	0
300	3	$\delta = 0.50$	0.425	98.86	93.72	99.94	99.94	0
400	3	$\delta = 0.50$	0.425	99.88	97.97	100.00	100.00	0
500	3	$\delta = 0.50$	0.425	99.98	99.40	100.00	100.00	0

Table 8: Survival-only effect: estimated power by sample size and effect level

n	h	Effect	Exp. Δ	Wilcoxon	T-test	LRT	SPLRT	Max fail
50	0	$\pi_1 = 0.45$	0.00	4.83	4.86	5.48	5.64	0
100	0	$\pi_1 = 0.45$	0.00	5.28	5.24	5.38	5.49	0
150	0	$\pi_1 = 0.45$	0.00	4.95	5.02	5.18	5.24	0
200	0	$\pi_1 = 0.45$	0.00	5.02	5.01	4.96	5.00	0
300	0	$\pi_1 = 0.45$	0.00	4.69	4.72	4.96	4.95	0
400	0	$\pi_1 = 0.45$	0.00	4.96	4.97	4.83	4.84	0
500	0	$\pi_1 = 0.45$	0.00	5.02	5.01	4.94	4.95	0
50	1	$\pi_1 = 0.50$	0.15	7.70	7.74	7.74	7.91	0
100	1	$\pi_1 = 0.50$	0.15	9.90	9.76	9.04	9.04	0
150	1	$\pi_1 = 0.50$	0.15	13.13	12.89	11.96	12.01	0
200	1	$\pi_1 = 0.50$	0.15	15.28	14.79	13.08	13.09	0
300	1	$\pi_1 = 0.50$	0.15	20.23	19.51	17.86	17.88	0
400	1	$\pi_1 = 0.50$	0.15	26.26	25.10	22.88	22.90	0
500	1	$\pi_1 = 0.50$	0.15	31.61	30.40	27.64	27.64	0
50	2	$\pi_1 = 0.55$	0.30	14.73	14.51	13.93	14.15	0
100	2	$\pi_1 = 0.55$	0.30	25.27	24.69	23.11	23.11	0
150	2	$\pi_1 = 0.55$	0.30	35.54	34.42	31.94	31.98	0
200	2	$\pi_1 = 0.55$	0.30	45.40	43.97	41.52	41.54	0
300	2	$\pi_1 = 0.55$	0.30	61.90	60.21	58.91	58.90	0
400	2	$\pi_1 = 0.55$	0.30	74.22	72.58	71.88	71.86	0
500	2	$\pi_1 = 0.55$	0.30	83.02	81.57	81.38	81.39	0
50	3	$\pi_1 = 0.60$	0.45	27.30	27.02	25.59	25.84	0
100	3	$\pi_1 = 0.60$	0.45	49.18	48.44	46.92	47.00	0
150	3	$\pi_1 = 0.60$	0.45	66.08	64.98	64.40	64.39	0
200	3	$\pi_1 = 0.60$	0.45	78.67	77.90	77.57	77.59	0
300	3	$\pi_1 = 0.60$	0.45	92.17	91.39	92.10	92.10	0
400	3	$\pi_1 = 0.60$	0.45	97.29	97.01	97.50	97.50	0
500	3	$\pi_1 = 0.60$	0.45	99.09	99.00	99.30	99.30	0

Table 9: Antagonistic cancellation: estimated power by sample size and effect level

n	h	Effect	Exp. Δ	Wilcoxon	T-test	LRT	SPLRT	Max fail
50	0	$\pi_1 = 0.500$ $\mu_0 + \mu_\delta = 3.000$	0	5.08	5.15	5.60	5.77	0
100	0	$\pi_1 = 0.500$ $\mu_0 + \mu_\delta = 3.000$	0	4.75	4.94	5.16	5.19	0
150	0	$\pi_1 = 0.500$ $\mu_0 + \mu_\delta = 3.000$	0	5.13	5.06	5.21	5.26	0
200	0	$\pi_1 = 0.500$ $\mu_0 + \mu_\delta = 3.000$	0	4.88	4.83	4.98	5.00	0

Table 9: Antagonistic cancellation: estimated power by sample size and effect level *(continued)*

n	h	Effect	Exp. Δ	Wilcoxon	T-test	LRT	SPLRT	Max fail
		$\pi_1 = 0.500$						
300	0		0	4.85	4.86	5.00	5.01	0
		$\mu_0 + \mu_\delta = 3.000$						
		$\pi_1 = 0.500$						
400	0		0	5.04	4.93	5.05	5.04	0
		$\mu_0 + \mu_\delta = 3.000$						
		$\pi_1 = 0.500$						
500	0		0	5.10	5.12	5.14	5.14	0
		$\mu_0 + \mu_\delta = 3.000$						
		$\pi_1 = 0.525$						
50	1		0	5.10	5.16	8.12	8.39	0
		$\mu_0 + \mu_\delta = 2.857$						
		$\pi_1 = 0.525$						
100	1		0	4.92	4.94	10.43	10.46	0
		$\mu_0 + \mu_\delta = 2.857$						
		$\pi_1 = 0.525$						
150	1		0	4.81	4.80	13.06	13.12	0
		$\mu_0 + \mu_\delta = 2.857$						
		$\pi_1 = 0.525$						
200	1		0	5.01	4.99	16.37	16.44	0
		$\mu_0 + \mu_\delta = 2.857$						
		$\pi_1 = 0.525$						
300	1		0	4.90	4.86	21.46	21.48	0
		$\mu_0 + \mu_\delta = 2.857$						
		$\pi_1 = 0.525$						
400	1		0	5.15	5.03	28.53	28.52	0
		$\mu_0 + \mu_\delta = 2.857$						
		$\pi_1 = 0.525$						
500	1		0	5.08	4.99	34.50	34.49	0
		$\mu_0 + \mu_\delta = 2.857$						
		$\pi_1 = 0.550$						
50	2		0	4.74	4.92	15.72	16.10	0
		$\mu_0 + \mu_\delta = 2.727$						
		$\pi_1 = 0.550$						
100	2		0	5.33	5.22	27.74	27.78	0
		$\mu_0 + \mu_\delta = 2.727$						
		$\pi_1 = 0.550$						
150	2		0	5.19	5.06	39.20	39.23	0
		$\mu_0 + \mu_\delta = 2.727$						
		$\pi_1 = 0.550$						
200	2		0	5.12	4.99	49.88	49.83	0
		$\mu_0 + \mu_\delta = 2.727$						
		$\pi_1 = 0.550$						
300	2		0	5.51	5.10	68.10	68.15	0
		$\mu_0 + \mu_\delta = 2.727$						
		$\pi_1 = 0.550$						
400	2		0	5.60	5.09	80.76	80.78	0
		$\mu_0 + \mu_\delta = 2.727$						
		$\pi_1 = 0.550$						
500	2		0	5.56	5.10	89.15	89.10	0
		$\mu_0 + \mu_\delta = 2.727$						
		$\pi_1 = 0.600$						
50	3		0	5.41	5.06	45.44	45.90	0
		$\mu_0 + \mu_\delta = 2.500$						
		$\pi_1 = 0.600$						
100	3		0	5.71	5.20	76.60	76.64	0
		$\mu_0 + \mu_\delta = 2.500$						
		$\pi_1 = 0.600$						
150	3		0	5.65	4.88	91.47	91.49	0
		$\mu_0 + \mu_\delta = 2.500$						
		$\pi_1 = 0.600$						
200	3		0	6.06	5.08	97.12	97.12	0
		$\mu_0 + \mu_\delta = 2.500$						
		$\pi_1 = 0.600$						
300	3		0	6.32	4.93	99.78	99.78	0
		$\mu_0 + \mu_\delta = 2.500$						

Table 9: Antagonistic cancellation: estimated power by sample size and effect level *(continued)*

n	h	Effect	Exp. Δ	Wilcoxon	T-test	LRT	SPLRT	Max fail
400	3	$\pi_1 = 0.600$ $\mu_0 + \mu_\delta = 2.500$	0	6.46	5.04	99.97	99.97	0
500	3	$\pi_1 = 0.600$ $\mu_0 + \mu_\delta = 2.500$	0	7.22	5.14	100.00	100.00	0

Table 10: Concordant joint improvement: estimated power by sample size and effect level

n	h	Effect	Exp. Δ	Wilcoxon	T-test	LRT	SPLRT	Max fail
50	0	$\pi_1 = 0.50$ $\mu_0 + \mu_\delta = 3.00$	0.000	5.00	5.05	5.72	5.97	0
100	0	$\pi_1 = 0.50$ $\mu_0 + \mu_\delta = 3.00$	0.000	5.05	5.08	5.13	5.18	0
150	0	$\pi_1 = 0.50$ $\mu_0 + \mu_\delta = 3.00$	0.000	4.94	5.02	5.12	5.18	0
200	0	$\pi_1 = 0.50$ $\mu_0 + \mu_\delta = 3.00$	0.000	4.99	4.90	5.02	5.04	0
300	0	$\pi_1 = 0.50$ $\mu_0 + \mu_\delta = 3.00$	0.000	4.96	4.96	5.13	5.11	0
400	0	$\pi_1 = 0.50$ $\mu_0 + \mu_\delta = 3.00$	0.000	5.24	5.19	5.46	5.47	0
500	0	$\pi_1 = 0.50$ $\mu_0 + \mu_\delta = 3.00$	0.000	5.02	5.19	5.05	5.01	0
50	1	$\pi_1 = 0.55$ $\mu_0 + \mu_\delta = 3.15$	0.233	10.04	10.37	9.84	10.07	0
100	1	$\pi_1 = 0.55$ $\mu_0 + \mu_\delta = 3.15$	0.233	15.71	16.25	14.39	14.47	0
150	1	$\pi_1 = 0.55$ $\mu_0 + \mu_\delta = 3.15$	0.233	21.30	21.88	19.48	19.54	0
200	1	$\pi_1 = 0.55$ $\mu_0 + \mu_\delta = 3.15$	0.233	26.52	27.40	24.43	24.40	0
300	1	$\pi_1 = 0.55$ $\mu_0 + \mu_\delta = 3.15$	0.233	37.70	38.72	34.90	34.93	0
400	1	$\pi_1 = 0.55$ $\mu_0 + \mu_\delta = 3.15$	0.233	48.14	49.66	45.59	45.53	0
500	1	$\pi_1 = 0.55$ $\mu_0 + \mu_\delta = 3.15$	0.233	56.48	57.87	54.23	54.19	0
50	2	$\pi_1 = 0.60$ $\mu_0 + \mu_\delta = 3.30$	0.480	27.45	28.01	25.71	26.04	0
100	2	$\pi_1 = 0.60$ $\mu_0 + \mu_\delta = 3.30$	0.480	48.84	49.53	46.02	46.10	0
150	2	$\pi_1 = 0.60$ $\mu_0 + \mu_\delta = 3.30$	0.480	66.29	66.94	64.03	63.99	0
200	2	$\pi_1 = 0.60$ $\mu_0 + \mu_\delta = 3.30$	0.480	78.16	78.66	76.47	76.44	0

Table 10: Concordant joint improvement: estimated power by sample size and effect level *(continued)*

n	h	Effect	Exp. Δ	Wilcoxon	T-test	LRT	SPLRT	Max fail
300	2	$\pi_1 = 0.60$ $\mu_0 + \mu_\delta = 3.30$	0.480	91.95	92.20	91.80	91.84	0
400	2	$\pi_1 = 0.60$ $\mu_0 + \mu_\delta = 3.30$	0.480	97.39	97.56	97.56	97.56	0
500	2	$\pi_1 = 0.60$ $\mu_0 + \mu_\delta = 3.30$	0.480	99.20	99.27	99.33	99.33	0
50	3	$\pi_1 = 0.65$ $\mu_0 + \mu_\delta = 3.45$	0.743	55.57	55.71	52.62	53.11	0
100	3	$\pi_1 = 0.65$ $\mu_0 + \mu_\delta = 3.45$	0.743	84.54	84.45	83.25	83.26	0
150	3	$\pi_1 = 0.65$ $\mu_0 + \mu_\delta = 3.45$	0.743	95.45	95.40	95.32	95.32	0
200	3	$\pi_1 = 0.65$ $\mu_0 + \mu_\delta = 3.45$	0.743	98.77	98.69	98.76	98.74	0
300	3	$\pi_1 = 0.65$ $\mu_0 + \mu_\delta = 3.45$	0.743	99.92	99.92	99.91	99.91	0
400	3	$\pi_1 = 0.65$ $\mu_0 + \mu_\delta = 3.45$	0.743	100.00	100.00	100.00	100.00	0
500	3	$\pi_1 = 0.65$ $\mu_0 + \mu_\delta = 3.45$	0.743	100.00	100.00	100.00	100.00	0

Table 11: Gamma shape-change with mean shift: estimated power by sample size and effect level

n	h	Effect	Exp. Δ	Wilcoxon	T-test	LRT	SPLRT	Max fail
50	0	$k_1 = 9.0$ $\mu_0 + \mu_\delta = 3.0$	0.00	5.06	5.18	5.58	5.80	0
100	0	$k_1 = 9.0$ $\mu_0 + \mu_\delta = 3.0$	0.00	4.86	4.93	5.04	5.15	0
150	0	$k_1 = 9.0$ $\mu_0 + \mu_\delta = 3.0$	0.00	4.91	4.96	5.16	5.19	0
200	0	$k_1 = 9.0$ $\mu_0 + \mu_\delta = 3.0$	0.00	4.90	4.91	5.23	5.28	0
300	0	$k_1 = 9.0$ $\mu_0 + \mu_\delta = 3.0$	0.00	5.04	4.98	5.10	5.11	0
400	0	$k_1 = 9.0$ $\mu_0 + \mu_\delta = 3.0$	0.00	4.85	4.76	4.87	4.88	0
500	0	$k_1 = 9.0$ $\mu_0 + \mu_\delta = 3.0$	0.00	4.98	4.80	5.03	5.02	0
50	1	$k_1 = 6.0$ $\mu_0 + \mu_\delta = 3.2$	0.13	5.96	6.63	8.91	9.80	0
100	1	$k_1 = 6.0$ $\mu_0 + \mu_\delta = 3.2$	0.13	6.20	7.92	12.72	13.58	0

Table 11: Gamma shape-change with mean shift: estimated power by sample size and effect level *(continued)*

n	h	Effect	Exp. Δ	Wilcoxon	T-test	LRT	SPLRT	Max fail
		$k_1 = 6.0$						
150	1		0.13	6.85	9.26	16.30	17.12	0
		$\mu_0 + \mu_\delta = 3.2$						
		$k_1 = 6.0$						
200	1		0.13	8.29	11.55	21.71	22.67	0
		$\mu_0 + \mu_\delta = 3.2$						
		$k_1 = 6.0$						
300	1		0.13	9.30	14.70	30.54	31.38	0
		$\mu_0 + \mu_\delta = 3.2$						
		$k_1 = 6.0$						
400	1		0.13	11.04	18.03	40.04	40.98	0
		$\mu_0 + \mu_\delta = 3.2$						
		$k_1 = 6.0$						
500	1		0.13	12.30	21.38	48.68	49.41	0
		$\mu_0 + \mu_\delta = 3.2$						
		$k_1 = 4.0$						
50	2		0.26	6.47	9.68	15.12	18.01	0
		$\mu_0 + \mu_\delta = 3.4$						
		$k_1 = 4.0$						
100	2		0.26	8.02	14.93	27.95	31.22	0
		$\mu_0 + \mu_\delta = 3.4$						
		$k_1 = 4.0$						
150	2		0.26	9.87	20.93	41.44	44.35	0
		$\mu_0 + \mu_\delta = 3.4$						
		$k_1 = 4.0$						
200	2		0.26	11.56	26.98	52.85	55.68	0
		$\mu_0 + \mu_\delta = 3.4$						
		$k_1 = 4.0$						
300	2		0.26	14.77	38.62	72.55	74.50	0
		$\mu_0 + \mu_\delta = 3.4$						
		$k_1 = 4.0$						
400	2		0.26	18.50	48.62	84.90	86.06	0
		$\mu_0 + \mu_\delta = 3.4$						
		$k_1 = 4.0$						
500	2		0.26	22.02	58.51	92.46	93.10	0
		$\mu_0 + \mu_\delta = 3.4$						
		$k_1 = 2.5$						
50	3		0.39	6.28	12.90	18.73	24.58	0
		$\mu_0 + \mu_\delta = 3.6$						
		$k_1 = 2.5$						
100	3		0.39	7.42	23.42	37.29	43.65	0
		$\mu_0 + \mu_\delta = 3.6$						
		$k_1 = 2.5$						
150	3		0.39	8.97	34.56	56.16	61.55	0
		$\mu_0 + \mu_\delta = 3.6$						
		$k_1 = 2.5$						
200	3		0.39	10.05	44.68	70.41	74.56	0
		$\mu_0 + \mu_\delta = 3.6$						
		$k_1 = 2.5$						
300	3		0.39	12.63	61.43	88.36	90.35	0
		$\mu_0 + \mu_\delta = 3.6$						
		$k_1 = 2.5$						
400	3		0.39	15.41	74.24	95.70	96.42	0
		$\mu_0 + \mu_\delta = 3.6$						
		$k_1 = 2.5$						
500	3		0.39	17.37	83.40	98.56	98.84	0
		$\mu_0 + \mu_\delta = 3.6$						

Table 12: Rare-outlier contamination: estimated power by sample size and effect level

n	h	Effect	Exp. Δ	Wilcoxon	T-test	LRT	SPLRT	Max fail
50	0	$\delta = 0.00$	0.00	4.96	4.65	4.68	12.55	0
100	0	$\delta = 0.00$	0.00	4.82	4.88	5.02	6.58	0

Table 12: Rare-outlier contamination: estimated power by sample size and effect level *(continued)*

n	h	Effect	Exp. Δ	Wilcoxon	T-test	LRT	SPLRT	Max fail
150	0	$\delta = 0.00$	0.00	4.96	4.88	4.91	5.42	0
200	0	$\delta = 0.00$	0.00	5.08	4.89	5.01	5.27	0
300	0	$\delta = 0.00$	0.00	4.81	4.99	4.94	5.12	0
400	0	$\delta = 0.00$	0.00	5.00	4.77	4.88	4.98	0
500	0	$\delta = 0.00$	0.00	5.14	5.09	5.09	5.20	0
50	1	$\delta = 0.15$	0.12	22.17	6.10	7.68	15.14	0
100	1	$\delta = 0.15$	0.12	39.28	7.18	8.56	9.24	0
150	1	$\delta = 0.15$	0.12	54.42	8.15	9.98	9.90	0
200	1	$\delta = 0.15$	0.12	66.69	9.82	11.68	11.40	0
300	1	$\delta = 0.15$	0.12	83.44	11.46	14.13	13.68	0
400	1	$\delta = 0.15$	0.12	92.36	14.04	17.96	17.37	0
500	1	$\delta = 0.15$	0.12	96.88	16.62	21.98	21.39	0
50	2	$\delta = 0.25$	0.20	49.92	8.55	13.35	17.63	0
100	2	$\delta = 0.25$	0.20	78.80	11.43	15.25	14.25	0
150	2	$\delta = 0.25$	0.20	92.20	14.73	19.55	17.95	0
200	2	$\delta = 0.25$	0.20	97.50	17.81	24.27	22.52	0
300	2	$\delta = 0.25$	0.20	99.80	23.56	33.51	31.62	0
400	2	$\delta = 0.25$	0.20	99.96	30.24	43.54	41.67	0
500	2	$\delta = 0.25$	0.20	100.00	36.08	51.66	49.88	0
50	3	$\delta = 0.35$	0.28	74.76	11.10	18.32	18.73	0
100	3	$\delta = 0.35$	0.28	96.21	17.47	25.78	22.36	0
150	3	$\delta = 0.35$	0.28	99.52	23.70	34.07	30.58	0
200	3	$\delta = 0.35$	0.28	99.96	29.42	42.66	39.19	0
300	3	$\delta = 0.35$	0.28	100.00	40.94	59.25	56.05	0
400	3	$\delta = 0.35$	0.28	100.00	51.72	71.84	69.16	0
500	3	$\delta = 0.35$	0.28	100.00	60.11	81.16	79.27	0

Table 13: Scenario-matched null calibration at effect level $h = 0$

Scenario	n	Wilcoxon	T-test	LRT	SPLRT	Max fail
S1	50	4.94	5.03	5.32	5.37	0
S1	100	4.72	4.73	5.19	5.21	0
S1	150	4.98	5.02	5.10	5.11	0
S1	200	5.16	5.09	5.28	5.30	0
S1	300	5.00	5.00	4.94	4.94	0
S1	400	4.95	4.87	4.95	4.94	0
S1	500	4.68	4.71	4.78	4.78	0
S2	50	4.83	4.86	5.48	5.64	0
S2	100	5.28	5.24	5.38	5.49	0
S2	150	4.95	5.02	5.18	5.24	0
S2	200	5.02	5.01	4.96	5.00	0
S2	300	4.69	4.72	4.96	4.95	0
S2	400	4.96	4.97	4.83	4.84	0
S2	500	5.02	5.01	4.94	4.95	0
S3	50	5.08	5.15	5.60	5.77	0
S3	100	4.75	4.94	5.16	5.19	0
S3	150	5.13	5.06	5.21	5.26	0
S3	200	4.88	4.83	4.98	5.00	0
S3	300	4.85	4.86	5.00	5.01	0
S3	400	5.04	4.93	5.05	5.04	0
S3	500	5.10	5.12	5.14	5.14	0
S4	50	5.00	5.05	5.72	5.97	0
S4	100	5.05	5.08	5.13	5.18	0
S4	150	4.94	5.02	5.12	5.18	0
S4	200	4.99	4.90	5.02	5.04	0
S4	300	4.96	4.96	5.13	5.11	0
S4	400	5.24	5.19	5.46	5.47	0
S4	500	5.02	5.19	5.05	5.01	0
S5	50	5.06	5.18	5.58	5.80	0
S5	100	4.86	4.93	5.04	5.15	0
S5	150	4.91	4.96	5.16	5.19	0
S5	200	4.90	4.91	5.23	5.28	0
S5	300	5.04	4.98	5.10	5.11	0
S5	400	4.85	4.76	4.87	4.88	0

Table 13: Scenario-matched null calibration at effect level $h = 0$ *(continued)*

Scenario	n	Wilcoxon	T-test	LRT	SPLRT	Max fail
S5	500	4.98	4.80	5.03	5.02	0
S6	50	4.96	4.65	4.68	12.55	0
S6	100	4.82	4.88	5.02	6.58	0
S6	150	4.96	4.88	4.91	5.42	0
S6	200	5.08	4.89	5.01	5.27	0
S6	300	4.81	4.99	4.94	5.12	0
S6	400	5.00	4.77	4.88	4.98	0
S6	500	5.14	5.09	5.09	5.20	0

B Supplementary grid approximation for Delta-scale intervals

This appendix describes a finite-grid approximation to the likelihood-based Δ intervals defined in Section 2. The construction uses the joint likelihood-ratio surface $\mathcal{W}_n$ in Equation (16) and the profiled numerical contrast map $g_n^{(e_0)}$ in Equation (17).

Choose a finite rectangular grid of component contrasts,

$$\mathcal{G}_n = \{(m_a, b_b) : a = 1, \dots, A,\ b = 1, \dots, B\} \subset \mathbb{R}^2,$$

where m_a indexes candidate values of μ_δ and b_b indexes candidate values of β_δ. At each grid point compute

$$W_{ab} = \mathcal{W}_n(m_a, b_b), \qquad d_{ab} = g_n^{(e_0)}(m_a, b_b).$$

For the projected interval in Equation (20), retain the grid points inside the two-dimensional likelihood-ratio region,

$$\mathcal{A}_{2,n} = \left\{(a, b) : W_{ab} \le \chi^2_{2,1-\alpha}\right\}.$$

When $\mathcal{A}_{2,n}$ is non-empty, the grid approximation is

$$\widehat{\mathrm{CI}}^{\Delta,\mathrm{proj,grid}}_{1-\alpha} = \left[\min_{(a,b)\in\mathcal{A}_{2,n}} d_{ab}, \max_{(a,b)\in\mathcal{A}_{2,n}} d_{ab}\right].$$

For the profile interval in Equation (24), retain instead the grid points below the one-degree-of-freedom likelihood-ratio threshold,

$$\mathcal{A}_{1,n} = \left\{(a, b) : W_{ab} \le \chi^2_{1,1-\alpha}\right\}.$$

When $\mathcal{A}_{1,n}$ is non-empty, the grid approximation is

$$\widehat{\mathrm{CI}}^{\Delta,\mathrm{prof,grid}}_{1-\alpha} = \left[\min_{(a,b)\in\mathcal{A}_{1,n}} d_{ab}, \max_{(a,b)\in\mathcal{A}_{1,n}} d_{ab}\right].$$

Thus the projected and profile approximations use the same evaluated pairs (W_{ab}, d_{ab}) and differ only in the likelihood-ratio cutoff used to accept grid points. The approximation is a discretized image of the corresponding likelihood-ratio region; its accuracy depends on the grid resolution and on whether the grid covers the relevant part of the likelihood-ratio surface. In computation, the rectangular grid bounds are enlarged if an accepted point lies on the boundary of the evaluated rectangle. This expansion is repeated until the accepted set is separated from the boundary, or until a prespecified expansion limit is reached.

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